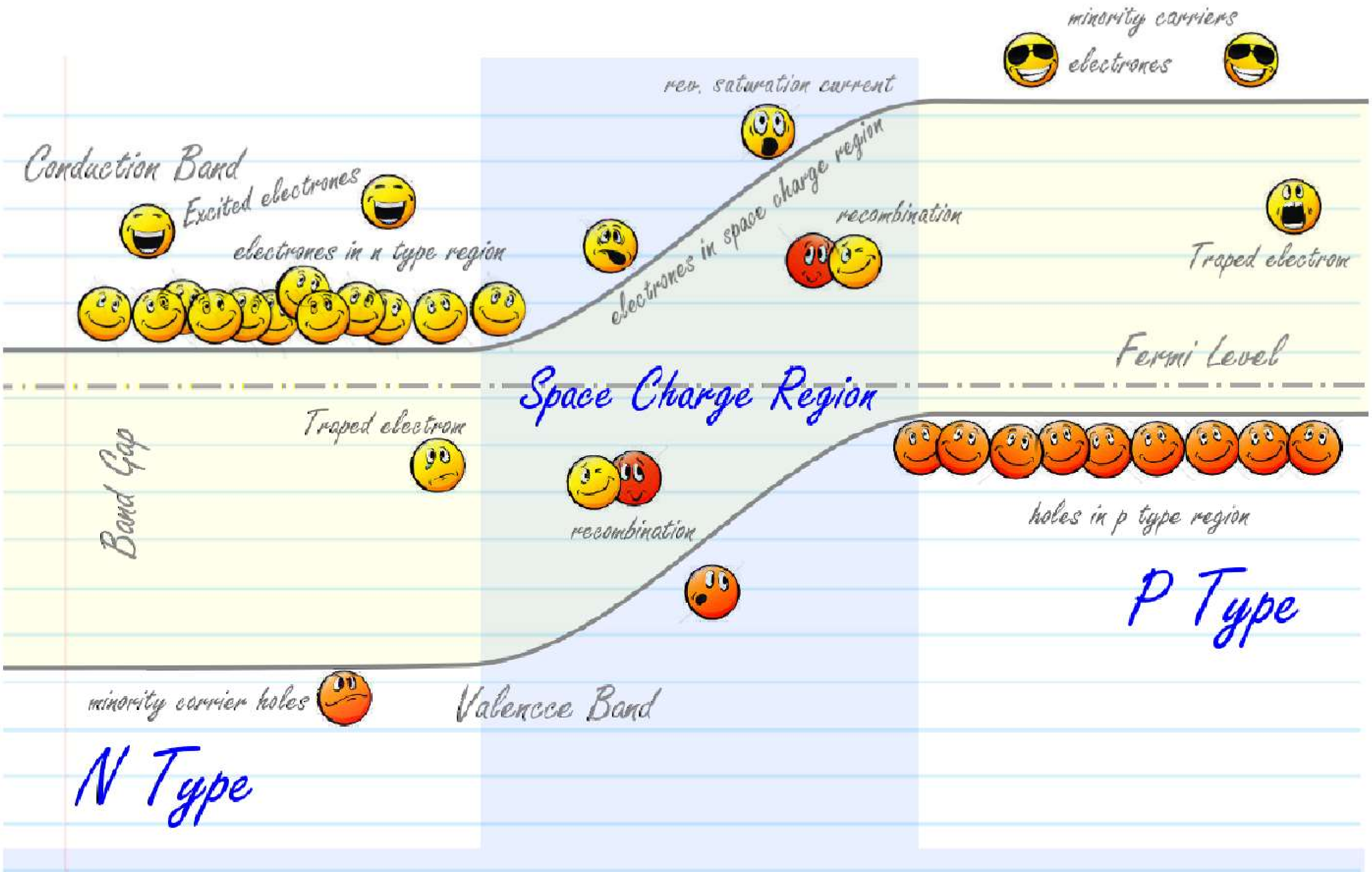
PN Diode I-V



minority carriers
electrones
rev. saturation current
Conduction Band
Excited electrones
electrones in space charge region
recombination
electrones in n type region
Traped electrom
Fermi Level
Space Charge Region
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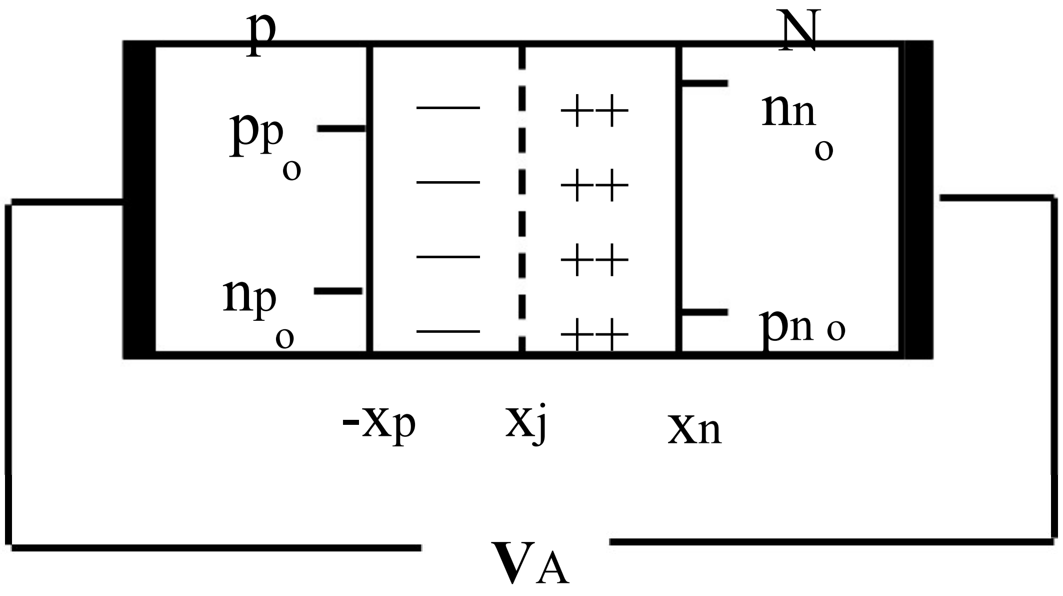
Band Gap
recombination
holes in p type region
P Type
minority carrier holes
Valencce Band
N Type

Junction Definitions



Define:

n_{no} = #electrons in N region in thermal equilibrium

n_{po} = #electrons in P region in thermal equilibrium

p_{no} = #holes in N region in thermal equilibrium

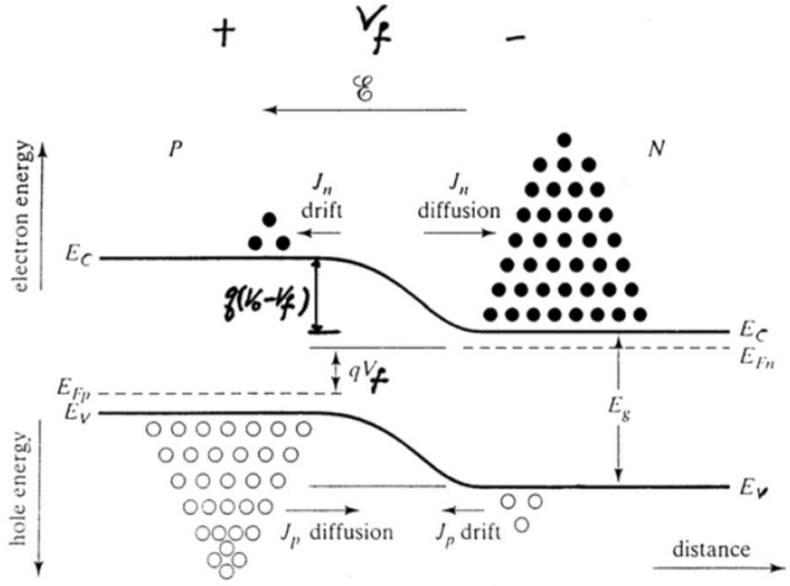
p_{po} = #holes in P region in thermal equilibrium

PN Junction under bias - Diffusion current

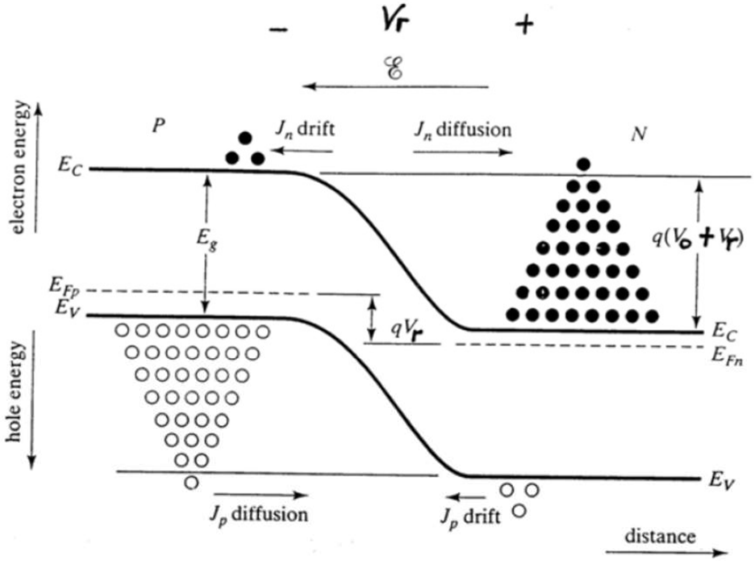
With zero bias, there is no diffusion current.

Under FB, the potential barrier for both electrons and holes is reduced to $\mathbf{q(V_o - V_f)}$.

Under RB, the potential barrier for both electrons and holes is increased to $\mathbf{q(V_o + V_r)}$.



Forward Bias



Reverse Bias

PN Junction under bias - Diffusion current

Under FB, electrons in the n-side CB have sufficient energy to diffuse from n to p over reduced potential barrier (same for holes in VB).

Thus, the electron (and hole) diffusion current increases sharply with FB.

Under RB, the increased barrier prevents electrons in the n-side CB to diffuse to p-side (same for holes in VB).

Thus, electron and hole diffusion currents, $\mathbf{J_n}$**(diff)** and $\mathbf{J_p}$**(diff)**

- <u>increase sharply under FB</u>
- <u>decrease sharply under RB</u>

PN Junction under bias - Drift current

Although the applied voltage **V** affects the built-in field ξ, $\mathbf{J_n}$**(drift)** and $\mathbf{J_p}$**(drift)** are in fact not affected by the strength of ξ and hence the applied bias **V**.

The carriers that are participating in the drift process are derived from the minority carriers in the neutral p and n regions just outside W.

Minority electrons at the p-side that wander into the depletion region will be swept by the ξ field across **W**, giving rise to $\mathbf{J_n}$**(drift)**.

Similarly minority holes at the n-side that wander into the depletion region will be swept by the ξ field across **W**, giving rise to $\mathbf{J_p}$**(drift)**.

PN Junction under bias - Drift current

As these are minority carriers, only limited number of them are available. In fact **J_n(drift)** and **J_p(drift)** are determined by the number of minority carriers entering W per sec, NOT by the electric field strength ξ in **W**.

In other words, the drift currents are limited not by how fast carriers are being swept across the depletion region, but rather by how many per second enter **W**.

Thus, **J_n(drift)** and **J_p(drift)** are insensitive to the applied bias **V**, and are only determined by the concentration of minority carriers in the neutral p and n regions just outside the depletion region **W**.

PN Junction under bias - Drift current

Supply of minority carriers on each side of the junction for participation in drift current is generated by thermal excitation of Electron-Hole pairs (EHP).

The EHP created near the junction on the p-side will provide the minority electron. If the EHP is generated within a diffusion length of the transition region, the electron can wander into the junction and be swept to the n-side by the electric field in **W**.

The resulting current is commonly called the generation current since its magnitude depends entirely on the rate of generation of EHPs and not the applied voltage.

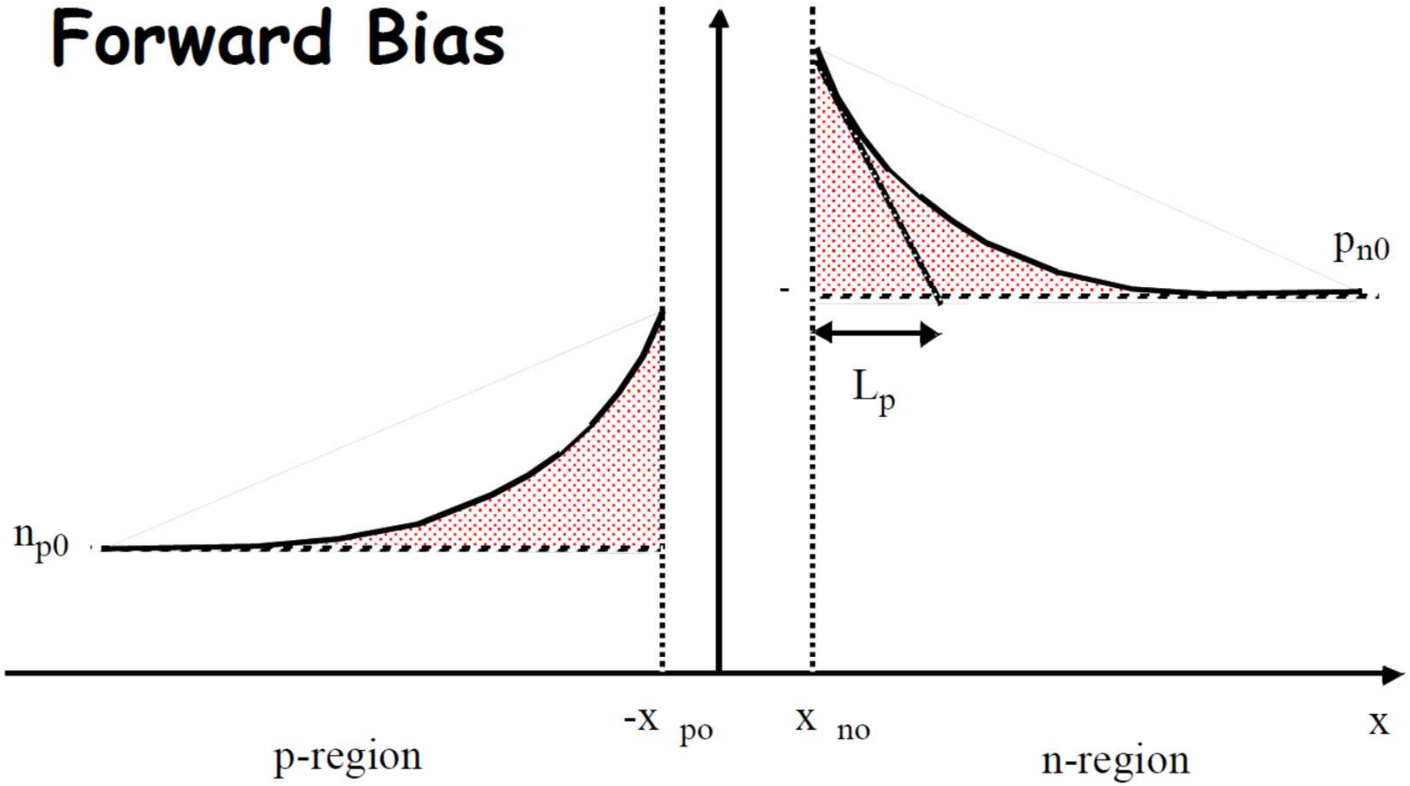
Similarly, we also have generation current due to holes on the n-side of the pn junction.

PN Junction under bias - Carrier concentration

At **thermal equilibrium**, the concentration of the minority carriers in the neutral p and n regions just outside **W** are given by $\mathbf{n_{po}} = \mathbf{n_i^2/N_a}$ (p-side) and $\mathbf{p_{no}} = \mathbf{n_i^2/N_d}$ (n-side). These equilibrium concentrations are maintained because the number of minority carriers lost per sec due to the drift process is just exactly balanced by the number of minority carriers being brought back by the diffusion process. (**J_p(diff) = J_p(drift), J_n(diff) = J_n(drift)**)

Under forward bias where diffusion of carriers is enhanced whereas drift of carriers remains unchanged (**J_p(diff) > J_p(drift), J_n(diff) > J_n(drift)**), the minority carrier concentrations in the neutral p and n regions outside W will be built-up above the equilibrium values ($\mathbf{n_{po}}$ and $\mathbf{p_{no}}$) due to the net diffusion of carriers. This is called **minority carrier injection.**

Forward Bias

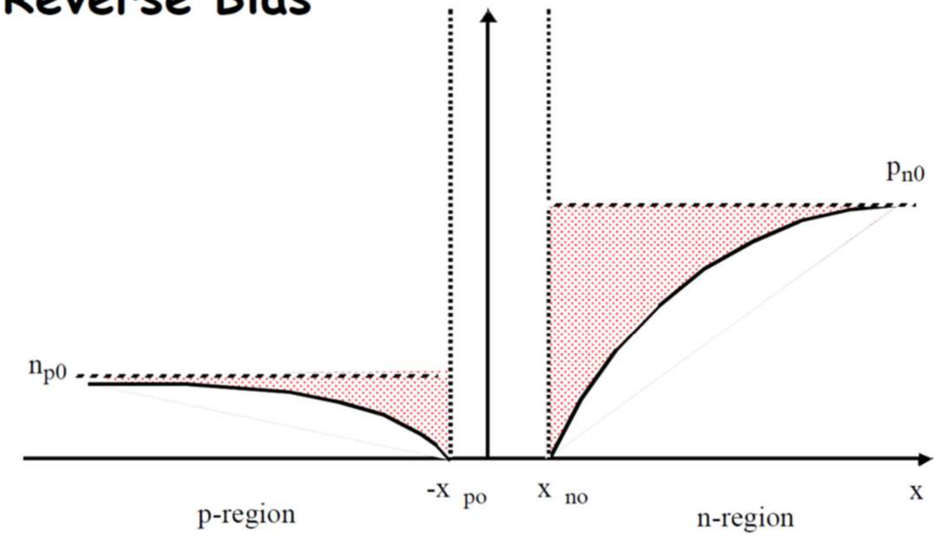


Minority carrier injection

PN Junction under bias - Carrier concentration

Under reverse bias, where diffusion of carriers is inhibited whereas drift of carriers remains unchanged (**J_p (diff) < J_p(drift)**, **J_n (diff) < J_n(drift)**), the minority carrier concentrations in the neutral p and n regions outside W will drop below the equilibrium values (**n_{po}** and **p_{no}**) due to the net drift of carriers. This is called **minority carrier extraction**.

Reverse Bias



Minority carrier extraction

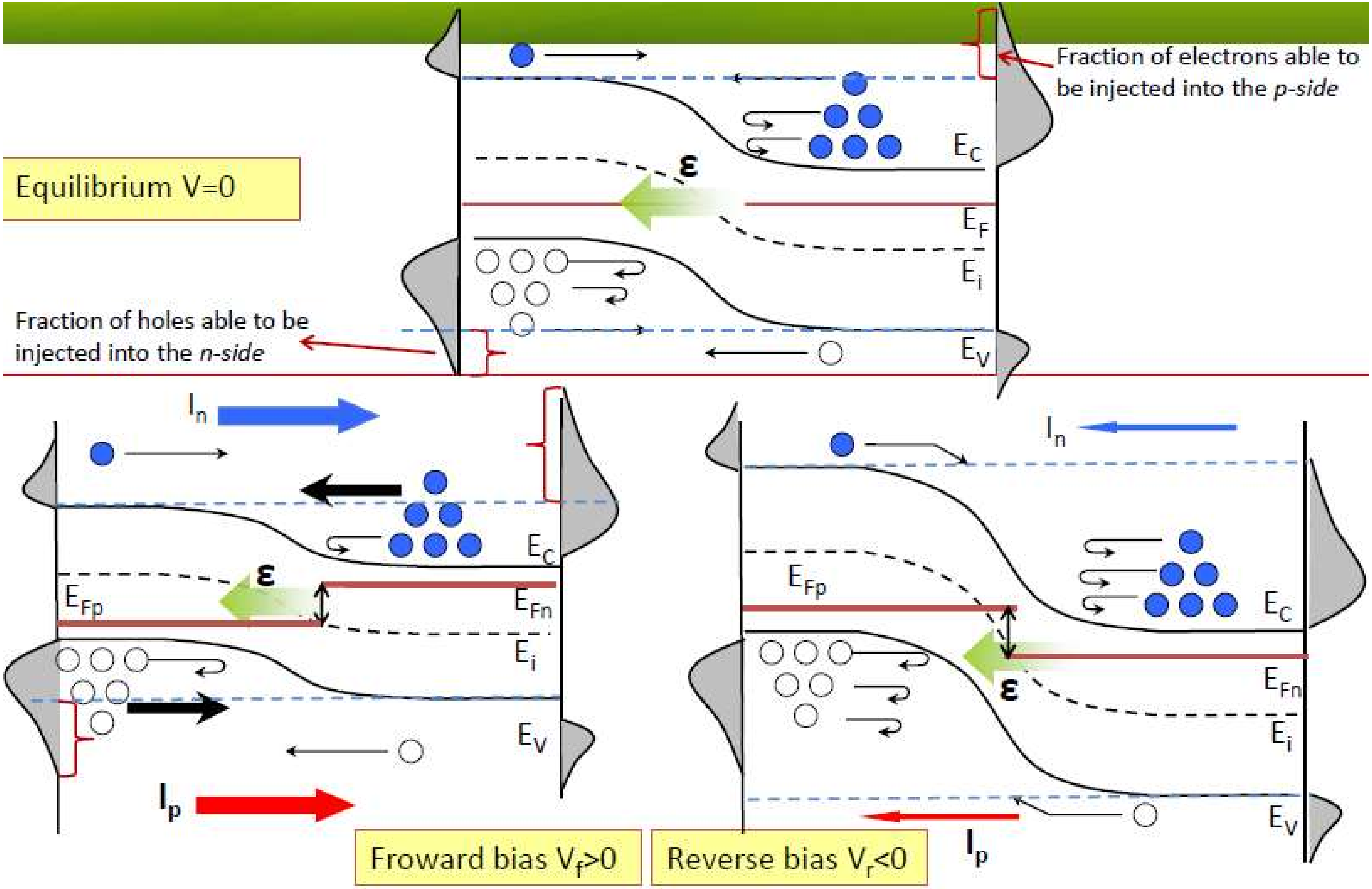
PN Junction under bias - summary

Under FB, **J_n(diff)** and **J_p(diff)** increase sharply due to lower potential barrier (**$q(V_o - V_f)$**). **J_n(drift)** and **J_p(drift)** remain unchanged at their thermal equilibrium values. Minority carrier injection takes place.

Net effect is a large diffusion current flowing from the p to n-side, which increases sharply with the applied voltage.

Under RB, **J_n(diff)** and **J_p(diff)** decrease sharply due to higher potential barrier (**$q(V_o + V_r)$**) and are negligible. **J_n(drift)** and **J_p(drift)** remain unchanged at their thermal equilibrium values. Minority carrier extraction takes place.

Net effect is a constant drift current flowing from n to p-side, which is insensitive to the applied **V**. This is the generation current (also called the reverse saturation current) of a pn junction.

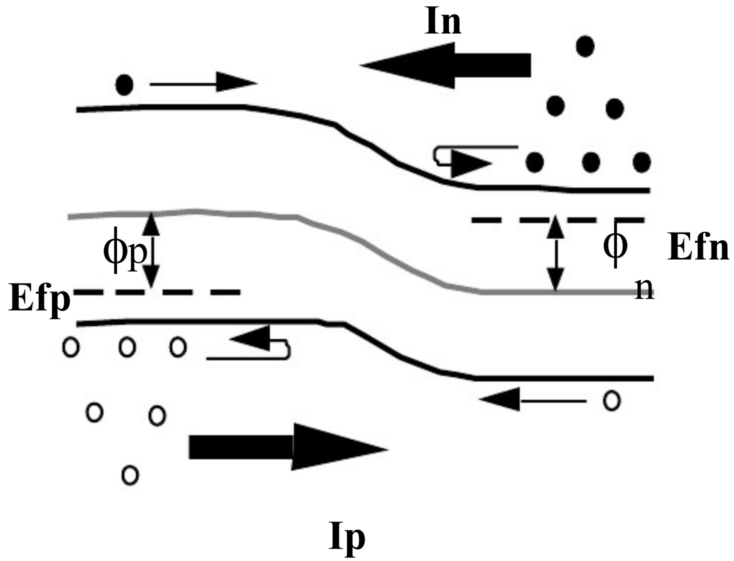


Equilibrium V=0
Fraction of electrons able to be injected into the *p-side*
Fraction of holes able to be injected into the *n-side*
E_C
E_F
E_i
E_V
ε
I_n
I_p
E_{Fp}
E_{Fn}
Froward bias V_f>0
Reverse bias V_r<0

Current Voltage Characteristics

Once electrons go from N $\rightarrow$ P they become minority carriers; similarly for holes from P $\rightarrow$ N. **Minority carrier behavior is of fundamental importance** in P-N junctions. The injection of minority carriers creates a concentration gradient near the edge of the depletion region. The minority carriers will therefore diffuse away from the depletion region (note: there is no electric field in the quasi-neutral region), and they will recombine if given sufficient time/distance.

The first thing needed is to relate the number of injected carriers to V_A.



Current Voltage Characteristics

$$n_{no} \approx N_D \text{ and } p_{no} = \frac{n_i^2}{N_D}$$

$$p_{po} \approx N_A \text{ and } n_{po} = \frac{n_i^2}{N_A}$$

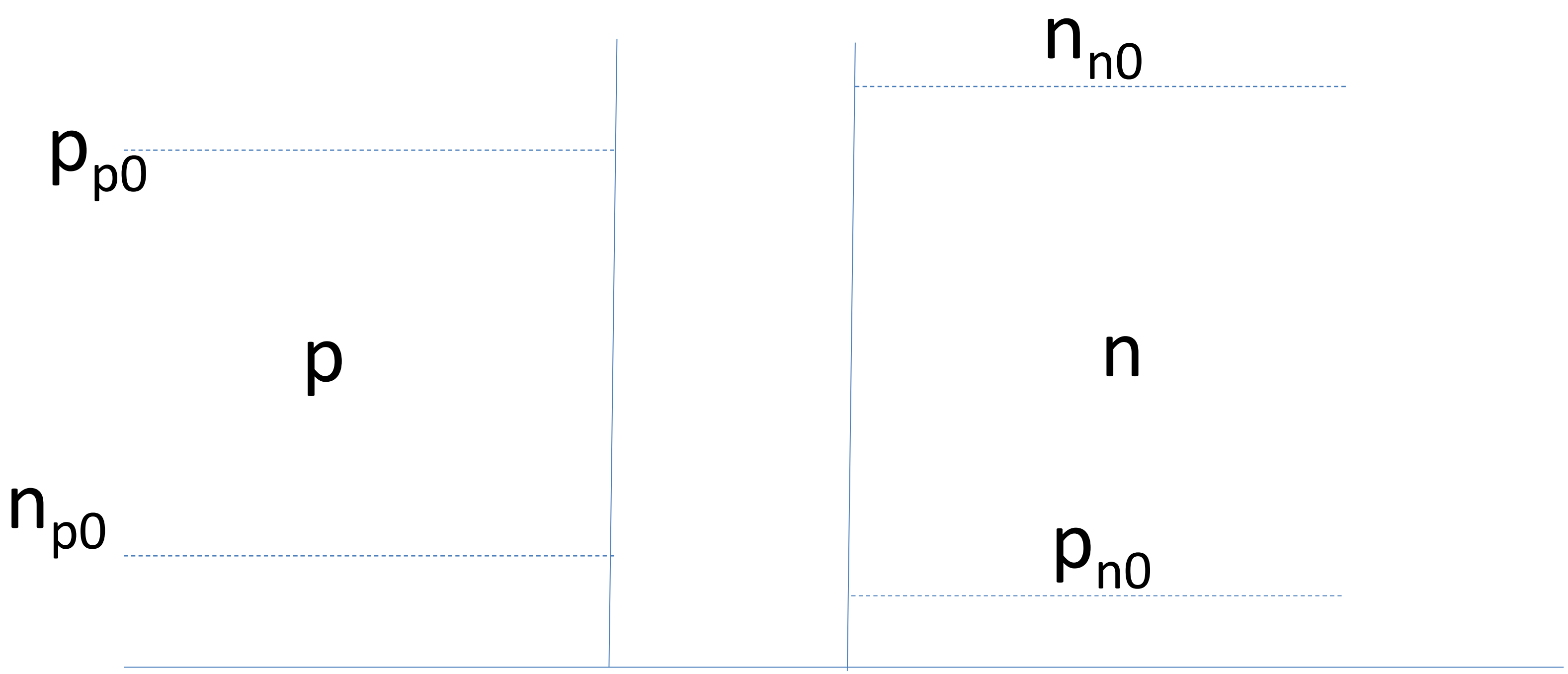
Complete donor and acceptor ionization

$$V_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} = \frac{kT}{q} \ln \frac{n_{no}}{n_{po}}$$

$$n_{no} = n_{po} e^{qV_{bi}/kT} \tag{20}$$

similarly, $p_{po} = p_{no} e^{qV_{bi}/kT}$ (21)

The minority carrier concentrations across the depletion region are related to the majority carrier concentrations on the other side by V_{bi}.



$$n_{p0} = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right)$$

$$p_{n0} = p_{p0} exp\left(\frac{-eV_{bi}}{kT}\right)$$

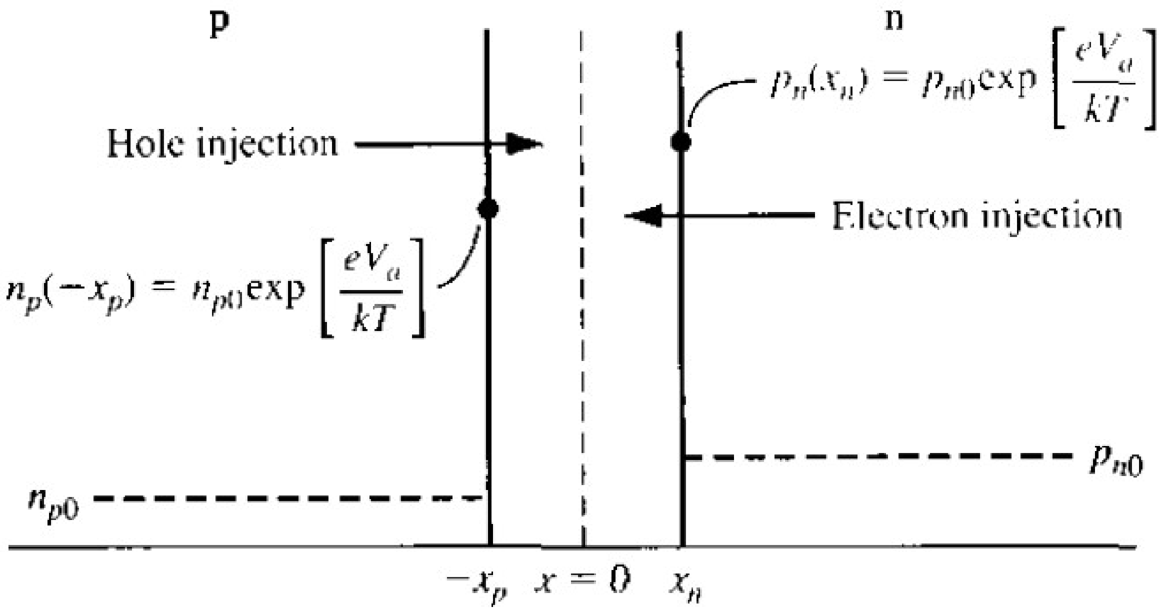
Under bias

$$n_{p0} = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right)$$

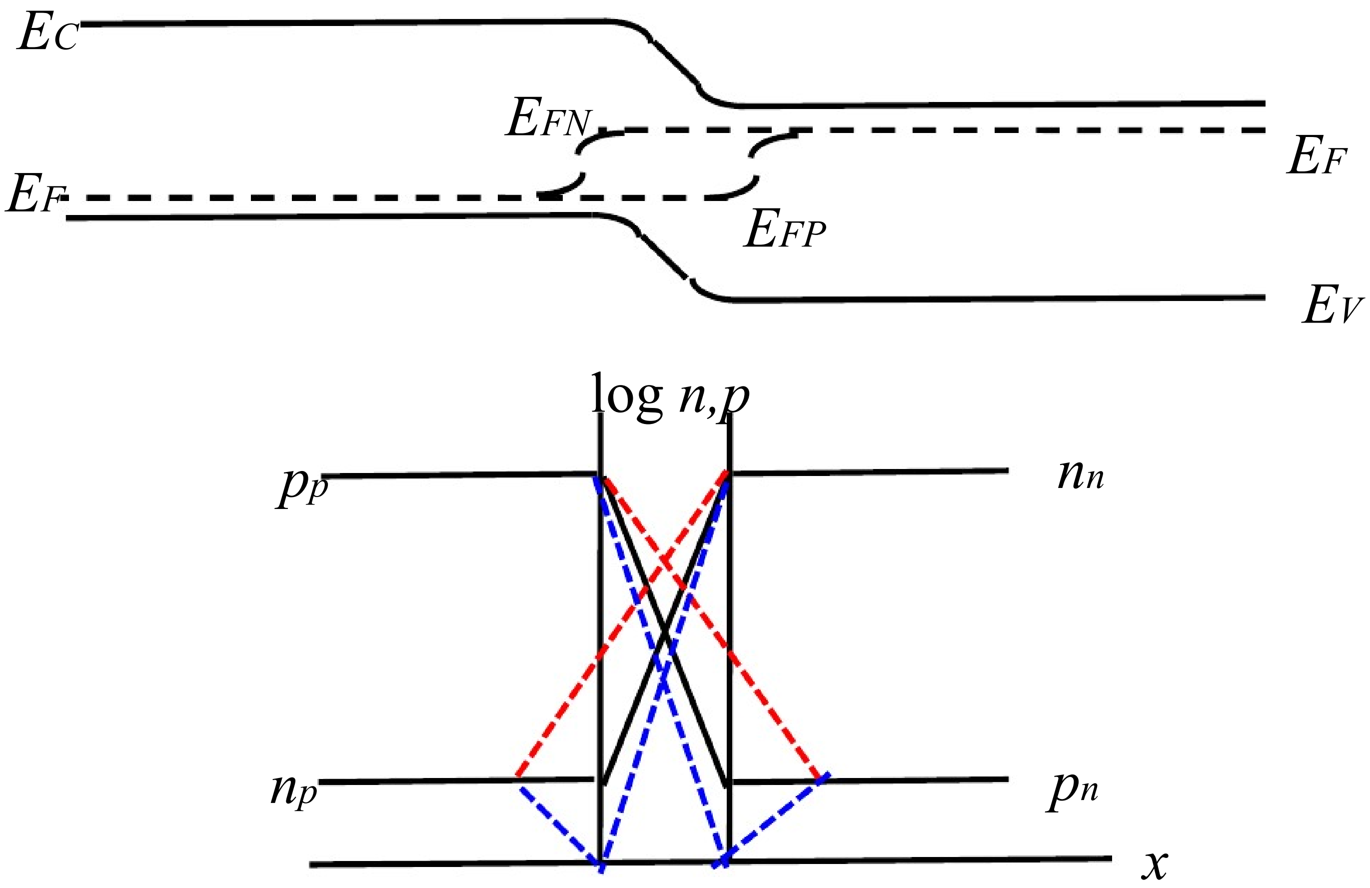
$$n_p = n_{n0} \exp\left(\frac{-e(V_{bi} - V_a)}{kT}\right) = n_{n0} \exp\left(\frac{-eV_{bi}}{kT}\right) \exp\left(\frac{+eV_a}{kT}\right)$$

$$\boxed{n_p = n_{p0} \exp\left(\frac{eV_a}{kT}\right)}$$

$$\boxed{p_n = p_{n0} \exp\left(\frac{eV_a}{kT}\right)}$$



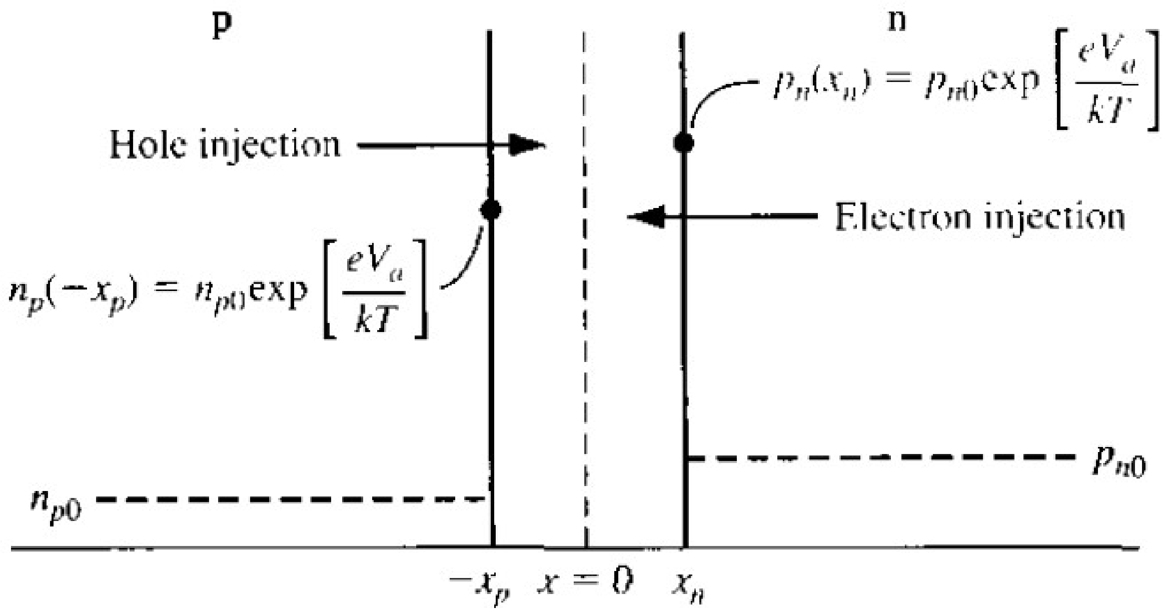
PN Junction Carrier Distributions



Forward
Reverse

Under bias

The carrier densities must be continuous across the "depletion" region to provide the carrier gradient for diffusion current. In the depletion approximation, we assume the carrier density to be zero, but this is only legitimate for solving Poisson's Equation



- The electron inject from n->p, become excess carrier
- Obey ambipolar transport theory

$$D_p \frac{\partial^2(\delta p_n)}{\partial x^2} - \mu_p \mathrm{E} \frac{\partial(\delta p_n)}{\partial x} + g' - \frac{\delta p_n}{\tau_{p0}} = \frac{\partial(\delta p_n)}{\partial t}$$

- Electric field inside the neutral n, p region is approximate zero

$$\frac{d^2(\delta p_n)}{dx^2} - \frac{\delta p_n}{L_p^2} = 0 \qquad (x > x_n) \qquad \frac{d^2(\delta n_p)}{dx^2} - \frac{\delta n_p}{L_n^2} = 0 \qquad (x < x_p)$$

$$L_p^2 = D_p \tau_{p0} \qquad\qquad L_n^2 = D_n \tau_{n0}.$$

- Solution to the continuity equation:

$$\delta p_n(x) = p_n(x) - p_{n0} = Ae^{x/L_p} + Be^{-x/L_p} \qquad (x \geq x_n)$$

$$\delta n_p(x) = n_p(x) - n_{p0} = Ce^{x/L_n} + De^{-x/L_n} \qquad (x \leq -x_p)$$

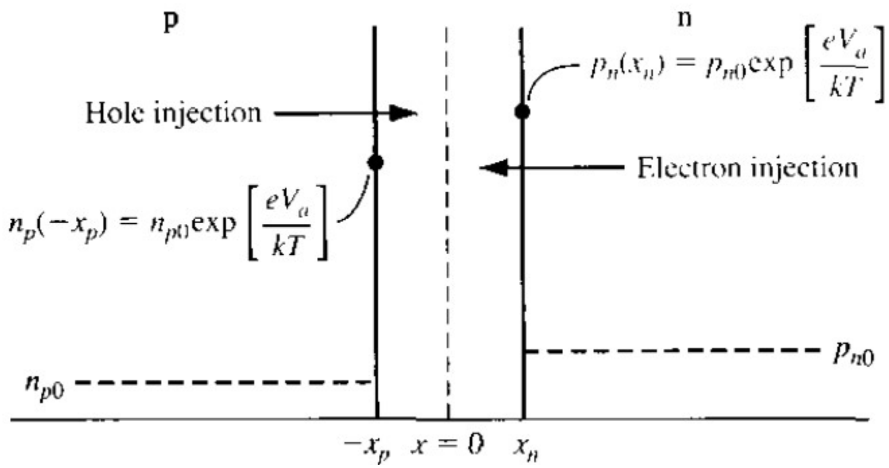
- Boundary conditions:

$$p_n(x_n) = p_{n0} \exp\left(\frac{eV_a}{kT}\right)$$

$$n_p(-x_p) = n_{p0} \exp\left(\frac{eV_a}{kT}\right)$$

$$p_n(x \to +\infty) = p_{n0}$$

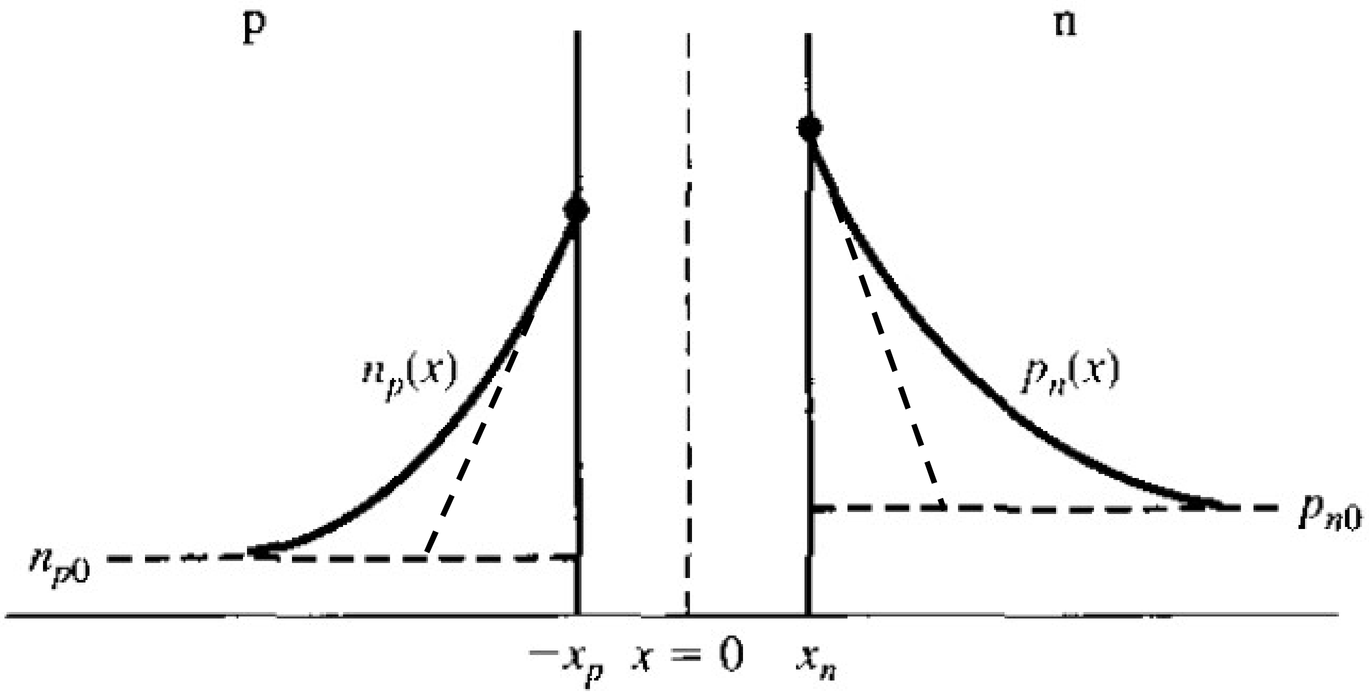
$$n_p(x \to -\infty) = n_{p0}$$



- Final solution:

$$\boxed{\delta p_n(x) = p_n(x) - p_{n0} = p_{n0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_n - x}{L_p}\right)}$$

$$\boxed{\delta n_p(x) = n_p(x) - n_{p0} = n_{p0}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_p + x}{L_n}\right)}$$



Holes are injected into the n-side and the value of this diffusion current is

$$I_p(x) = -eAD_p \frac{d(\delta p(x))}{dx} = eA\frac{D_p}{L_p}(\delta p(x)) \qquad x > x_n$$

The hole current injected into the n-side is given by the hole current at x = x_n (after using the value of δp(x =x_n) from the equation for δp)

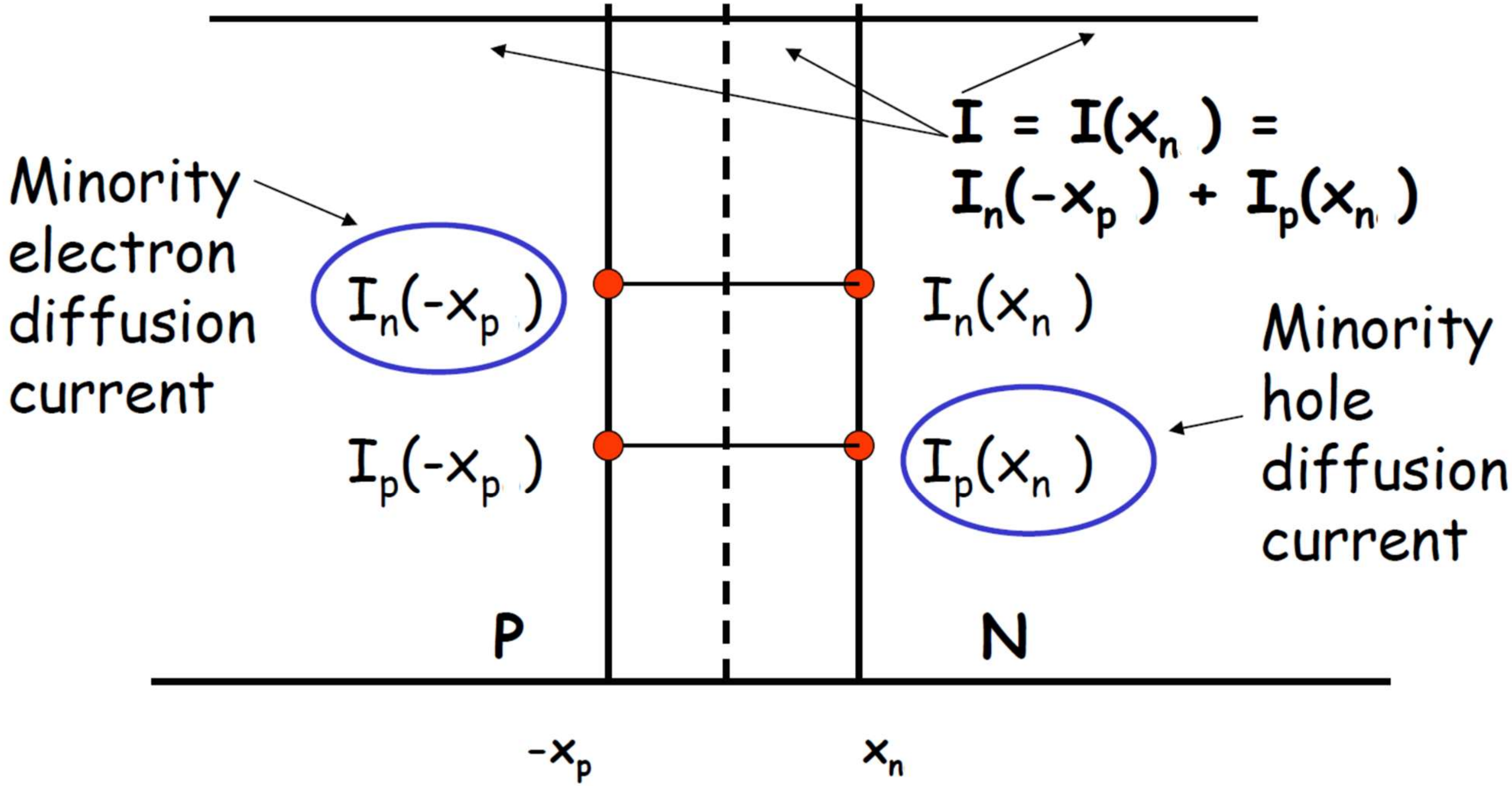
$$I_p(x_n) = e\frac{AD_p}{L_p} p_{n0}\left[\exp(eV / k_B T) - 1\right]$$

Using similar arguments, the total electron current injected across the depletion region into the p-side region is given by

$$I_n(-x_p) = e\frac{AD_n}{L_n} n_{p0}\left[\exp(eV / k_B T) - 1\right]$$

Here we will assume that the diode is ideal which essentially means there is n**o e-h recombination within the depletion region**. For the ***ideal diode*** the total current can be simply obtained by adding the hole current injected across x_n and electron current injected across $-x_p$.

PN Junction under bias – Diode Current



- The total current is constant throughout the device.
- A key point is that there is no recombination in the depletion region, so the **electron and hole currents are constant** across the depletion region.

The sum of the electron and hole currents in the depletion region, $I = I_p + I_n$ is given by $I_p(x_n)+I_n(-x_p)$ as the currents do not change in the depletion region due to the assumption of no generation - recombination. The ***ideal diode current*** is therefore

$$
\begin{aligned}
I(V) &= I_p(x_n) + I_n(-x_p) \\
&= eA\left[\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right]\left(e^{\frac{eV}{k_B T}} - 1\right) \\
&= I_0\left(e^{\frac{eV}{k_B T}} - 1\right)
\end{aligned}
$$

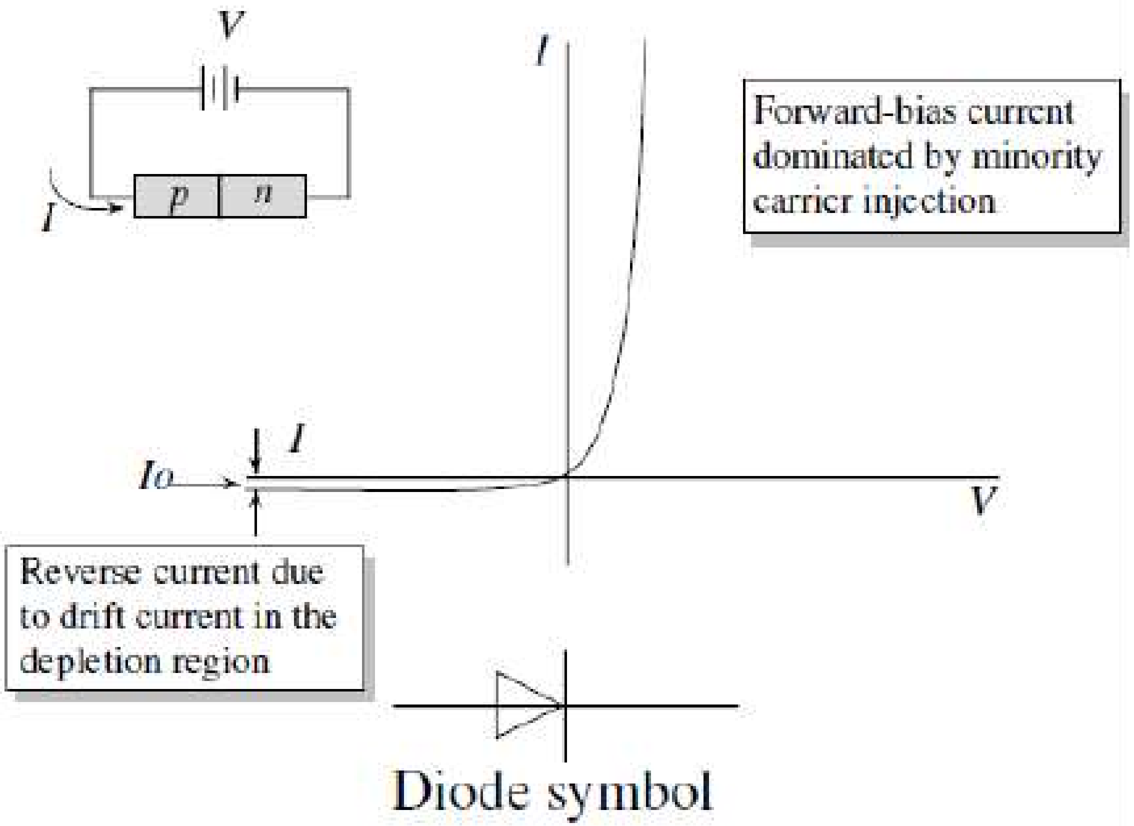
Under reverse bias, the current simply goes toward the value $-I_0$, where I_0 is the ***reverse saturation current***.

Notice that the diode current increases rapidly when a forward bias is applied and has a small value at negative bias. This gives the diode its rectification properties.

Show the reverse bias case

The rectifying properties of a diode are shown in the figure. The reverse current saturates to a value I_o given by the equation above. Since this value is quite small, the diode is essentially nonconducting.

On the other hand, when a positive bias is applied, the diode current increases exponentially and the diode becomes strongly conducting.



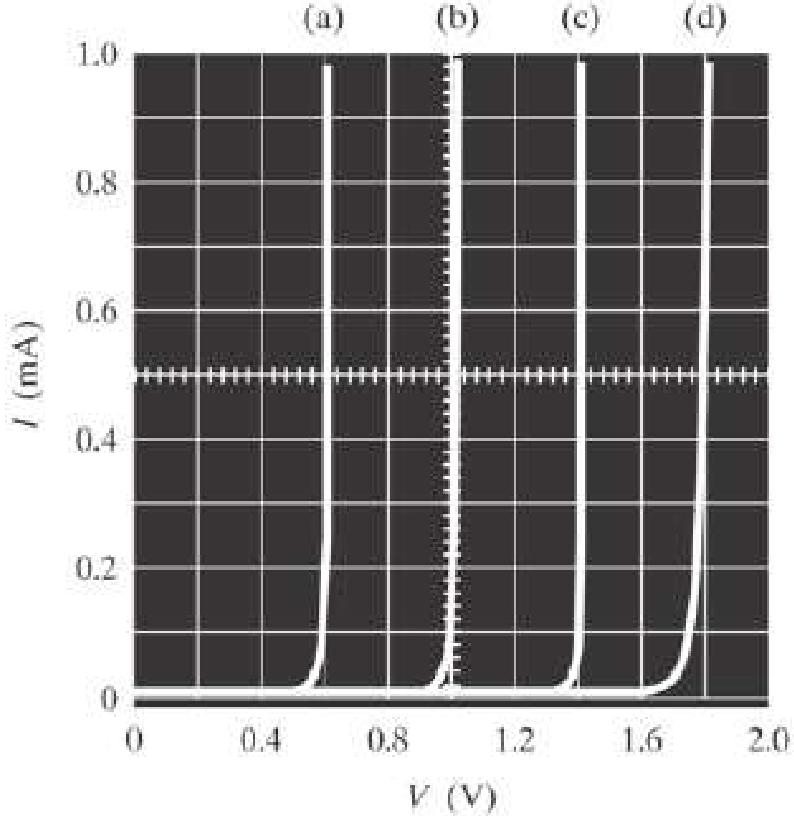
The forward bias voltage at which the diode current density becomes significant (~ 10^3 $A \cdot cm^{-2}$) is called the cut-in voltage. This voltage is~0.8V for Si diodes and ~1.2 V for GaAs diodes. The cut-in voltage is approximately 80 % of the material bandgap.

Why?

$$V_{bi} = \frac{kt}{q} ln \frac{N_d N_a}{n_i^2} = \frac{kt}{q} ln \frac{N_d N_a}{N_c N_v \exp(\frac{-E_g}{kt})} \approx \frac{E_g}{q}, for\ heavy\ doping$$

Hole injection and therefore the current is small if the forward bias V is much less than the V_0 (or E_{gap} for p^+ - n^+ junctions).

a) Ge Eg ≈ 0.7 eV
b) Si Eg ≈ 1.1 eV
c) GaAs Eg ≈ 1.4 eV
d) GaAsP Eg ≈ 1.9 eV

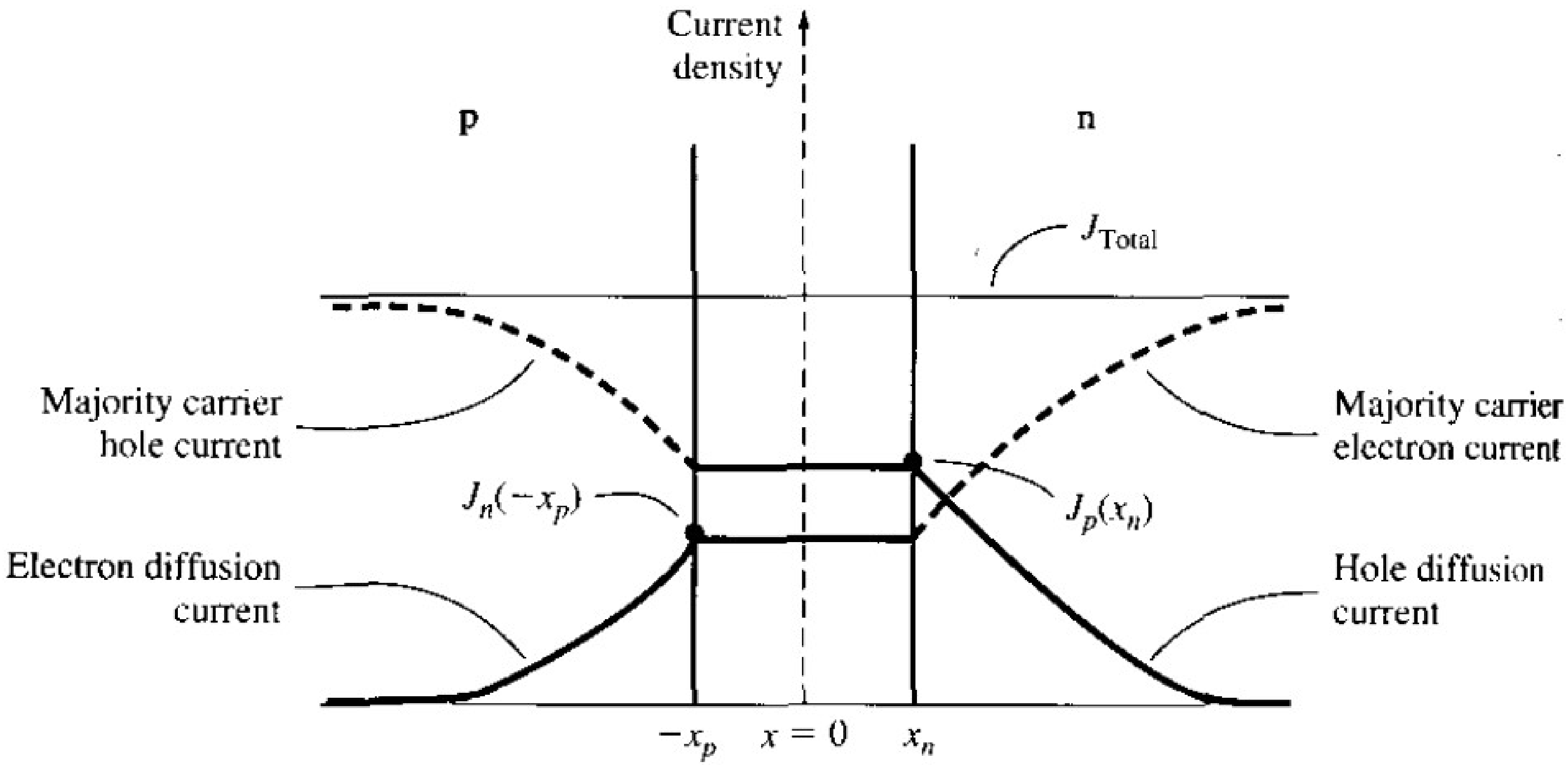


Diffusion current as a function of distance

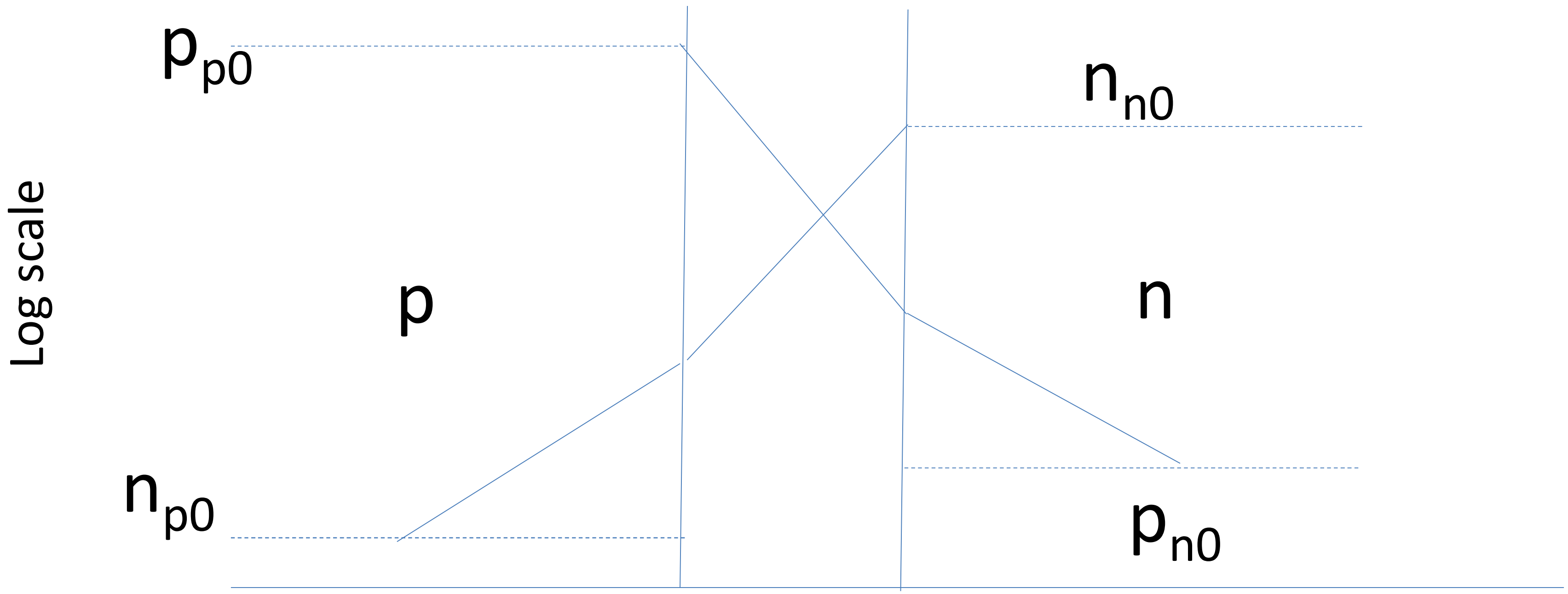
$$Jp(x) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_n - x}{L_p}\right) \qquad (x \geq x_n)$$

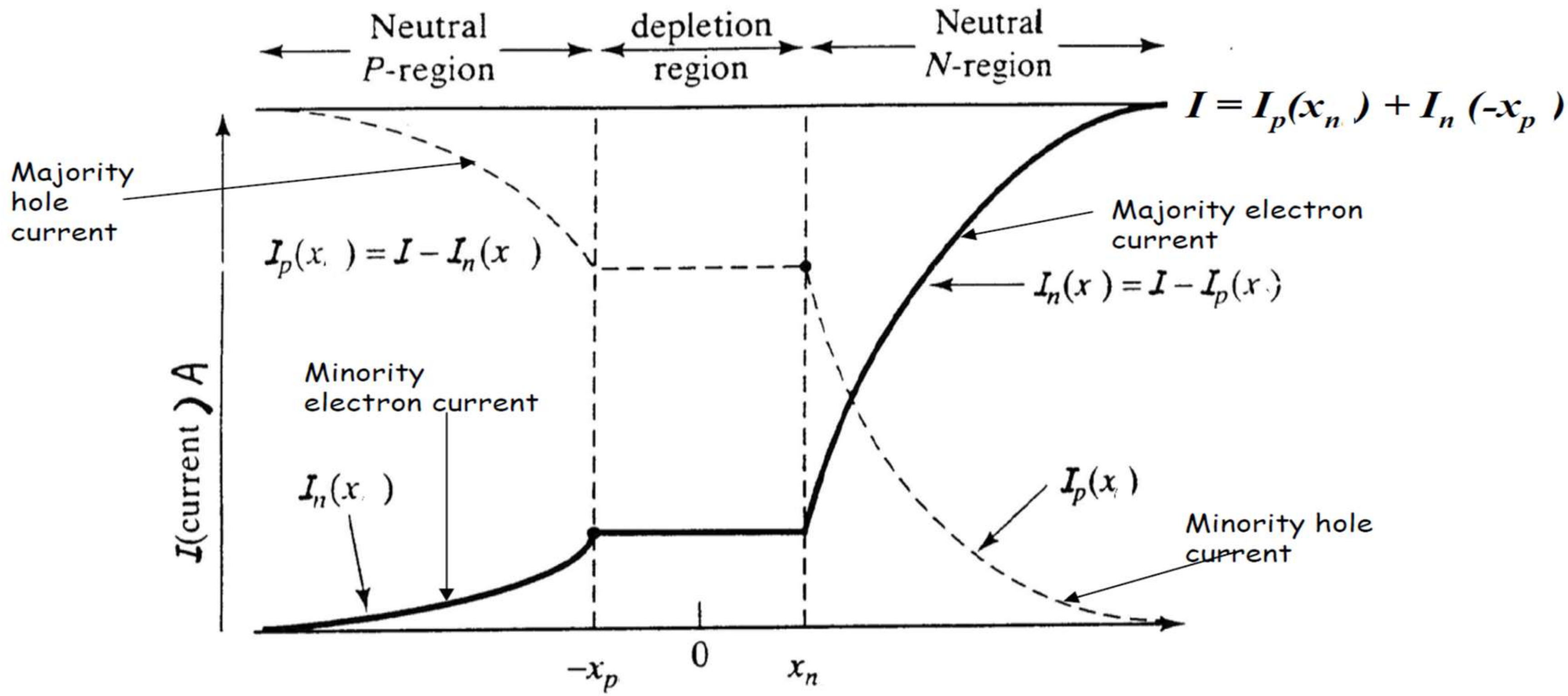
and

$$Jn(x) = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]\exp\left(\frac{x_p + x}{L_n}\right) \qquad (x \leq -x_p)$$

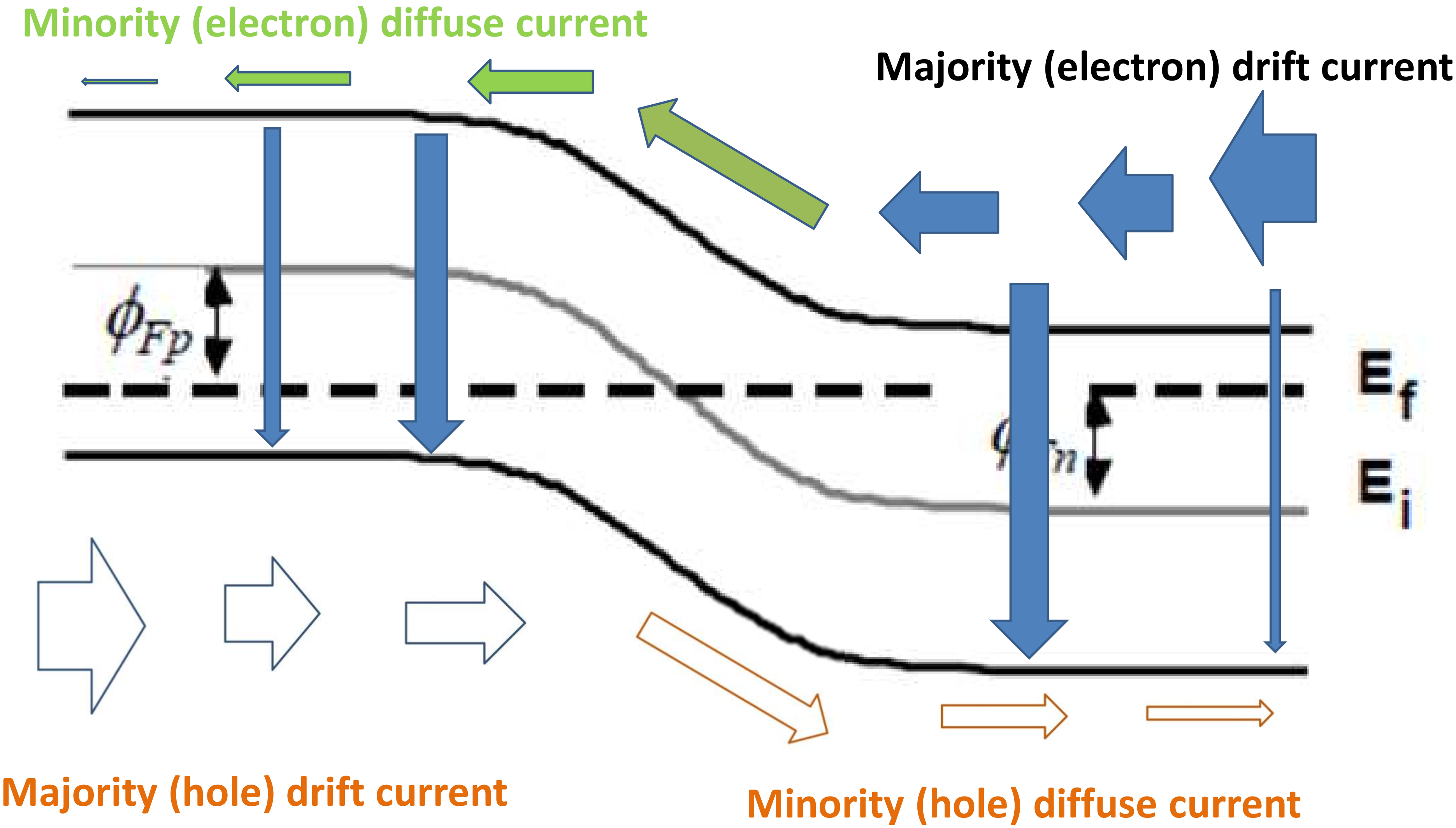


Carrier distribution under forward bias





PN Junction Under Forward Bias



Current Voltage Characteristics

1. The minority carrier densities decrease with distance due to recombination

2. From continuity of *I*, the majority currents must increase with distance, i.e., minority current is continuously transformed into majority carrier current by recombination.

The total current flowing across the junction is thus given by the sum of I_p and I_n at any point x, including the edges of the depletion region ($x = x_n$ and $x = -x_p$ respectively). Because this current results from the diffusion of minority carriers away from the depletion region, it is often called the *diffusion current*, I_{DIFF}.

$$I = qA\left(\frac{D_P}{L_P}\frac{n_i^2}{N_D} + \frac{D_N}{L_N}\frac{n_i^2}{N_A}\right)\left(e^{qV_A/kT} - 1\right) = I_o\left(e^{qV_A/kT} - 1\right) \qquad (30)$$

Current Voltage Characteristics

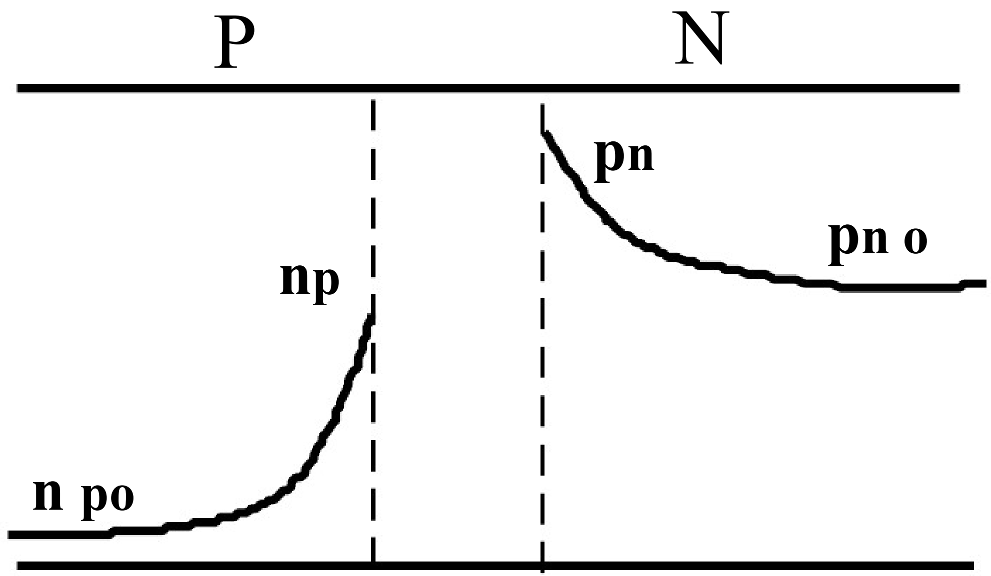
Where

$$I_o = qA\left(\frac{D_P}{L_P}\frac{n_i^2}{N_D} + \frac{D_N}{L_N}\frac{n_i^2}{N_A}\right) = qA\left(\frac{D_P p_{no}}{L_P} + \frac{D_N n_{po}}{L_N}\right) \quad (31)$$

This last result illustrates particularly well that it is the doping level on the **LIGHTLY DOPED** side of the junction that determines the current. In other words, majority carriers from the heavily doped side of the junction are injected as minority carriers into the lightly doped side. The resulting concentration gradient and greater number of minority carriers on the lightly doped side primarily determine the magnitude of the current.

Minority Carrier Distributions

For example: If N is lightly doped,



the current is mostly due to injection of holes into lightly doped n-side.

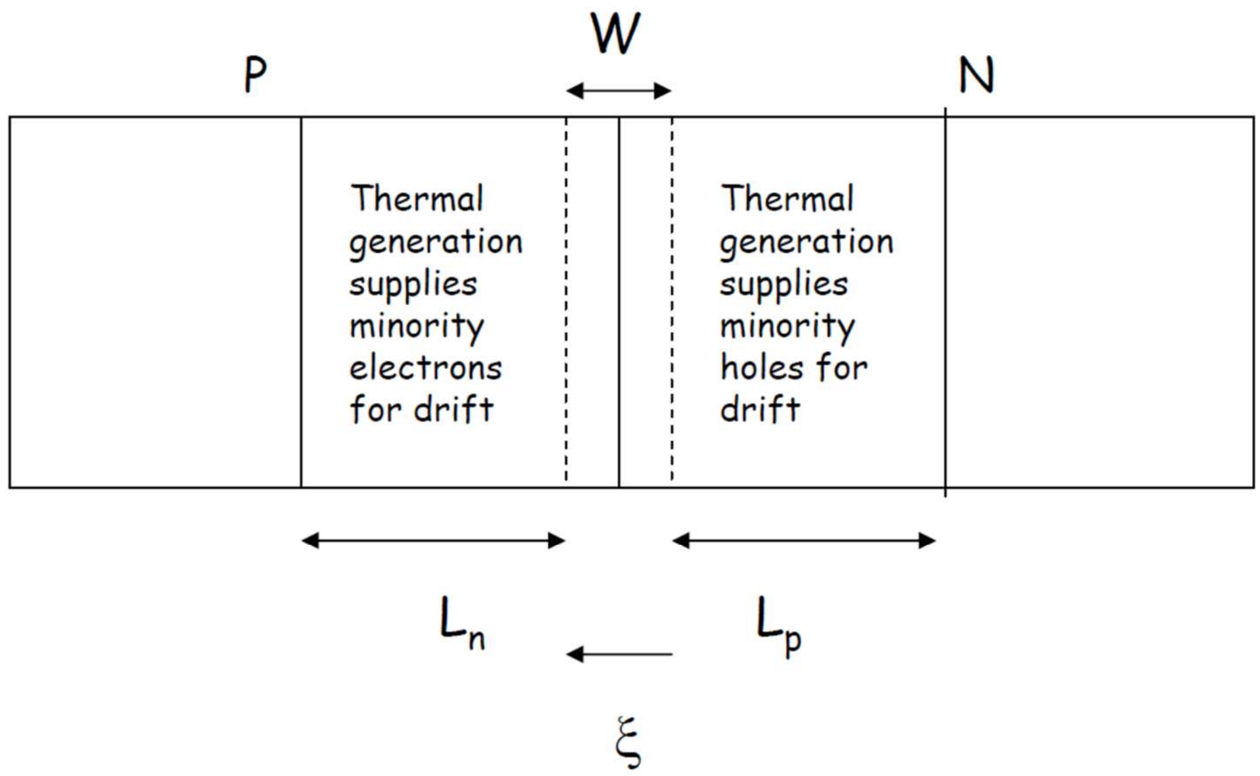
Diode Current-Minority and Majority carrier currents

Reverse Bias

- Diffusion process is inhibited because of high potential barrier. However, minority carrier holes are extracted from the n-region and minority carrier electrons are extracted from the p-region.
- The electric field in the depletion region is large enough to extract all the minority carriers near the junction.
- This flow of minority carriers across the junction leads to the reverse saturation or generation current.
- Minority carriers that are thermally generated within a minority carrier diffusion length ($\mathbf{L_n}$ at p-side and $\mathbf{L_p}$ at n-side) away from the edge of **W** will diffuse towards **W** and be swept across by the built-in field ξ, giving rise to the drift current.

Diode Current-Minority and Majority carrier currents

Reverse Bias



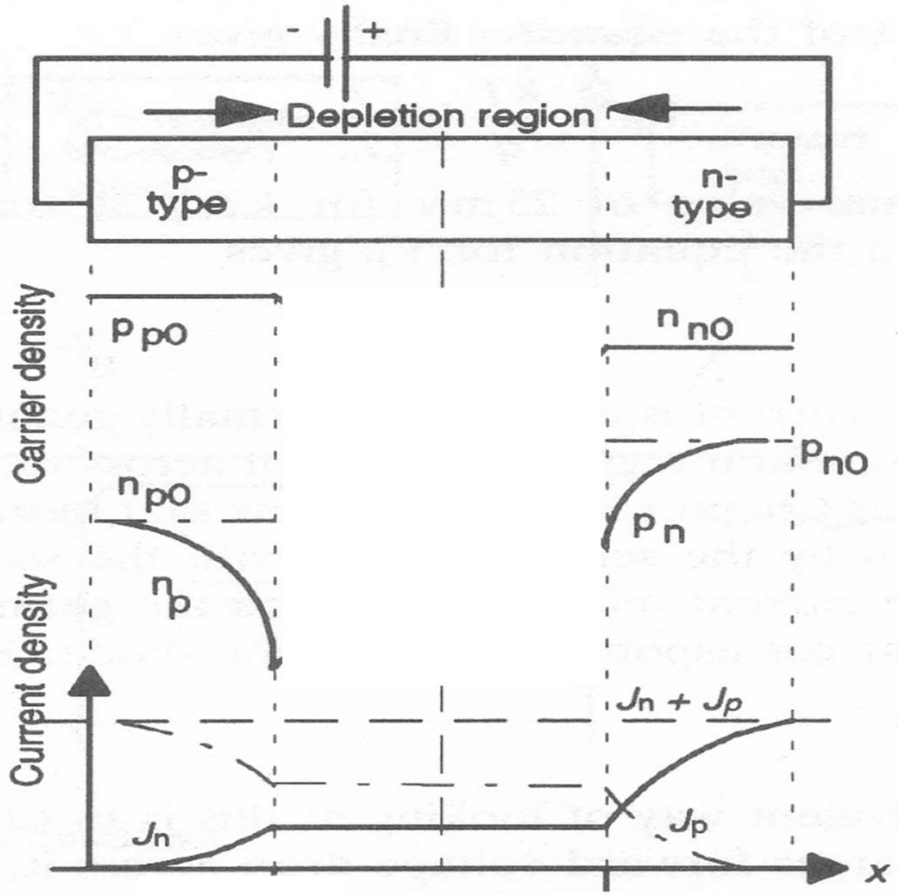
Diode Current-Minority and Majority carrier currents

Reverse Bias

- The name generation current arises from the fact that the thermal generation process continuously supplies the minority carriers that participate in the drift process, giving rise to J_n**(drift)** and J_p**(drift)**, the sum of which is the reverse saturation current I_o.
- The reverse saturation current will increase exponentially with temperature but is essentially independent of the reverse bias, provided junction breakdown will not occur.
- This current can also be increased by optical excitation, for example shining light near the metallurgical junction of a RBd pn junction will generate extra electron hole pairs and increase the minority carriers supply and hence I_o. This is the basis of using a RB pn junction as a photodetector.

Diode Current-Minority and Majority carrier currents

Reverse Bias



Diode Current-Minority and Majority carrier currents

Reverse Bias

- The minority carriers, upon crossing **W** by the drift process, will become majority carriers. They will then, together with the thermally generated majority carriers, be driven by the weak electric field in the neutral p and n regions and move towards both ends of the diode. Similar to the FB case, these weak fields are developed due to a small fraction of the applied reverse bias voltage $-V_r$ dropping across the p and n neutral regions.

- The circuit is completed by the majority electrons arriving at the end of the n region being driven out into the external circuit and enter into the p region to recombine with the majority holes that arrived at the end of the p region.

Diode – FB and RB currents

<u>What happens if we apply excessive **FB** to a diode?</u>

If we take a diode with a V_o = 0.6 V and apply a very large FB voltage, say 20 V. Do the bands bend by 19.4 V?

They do not. This is because we are looking at a diode in isolation only. In realistic situations, a diode circuit will have a current limiting resistor in series with the diode.

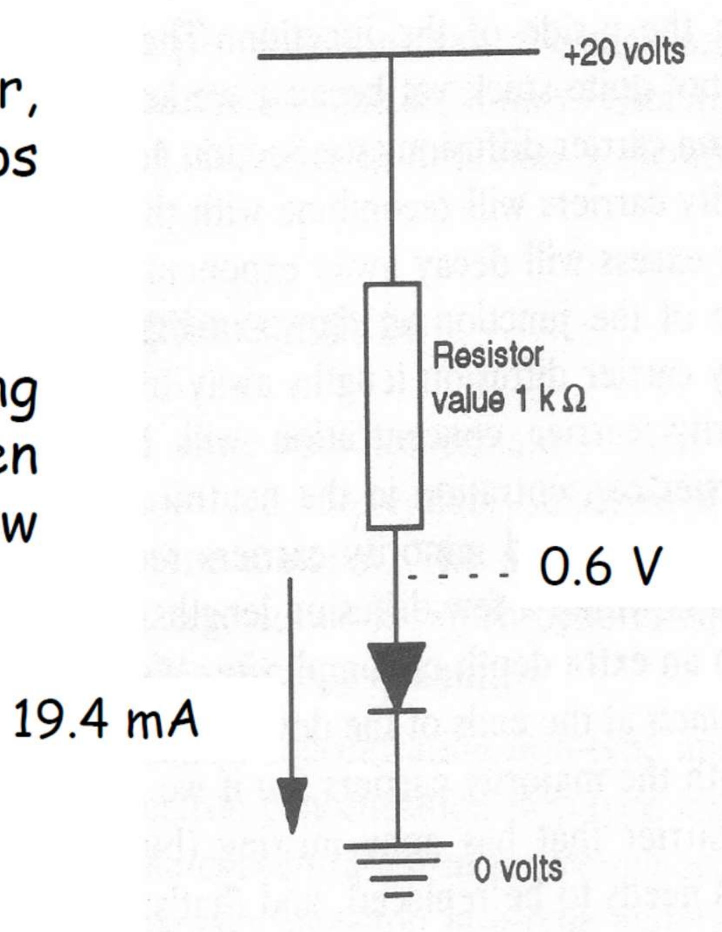
For the diode, 0.6V is sufficient to flatten the bands resulting in large and free flow of electrons and holes and there is virtually no limit to the diffusion currents (we are in a high-level injection regime).

This would simply lead to destruction of the device due to excessive current.

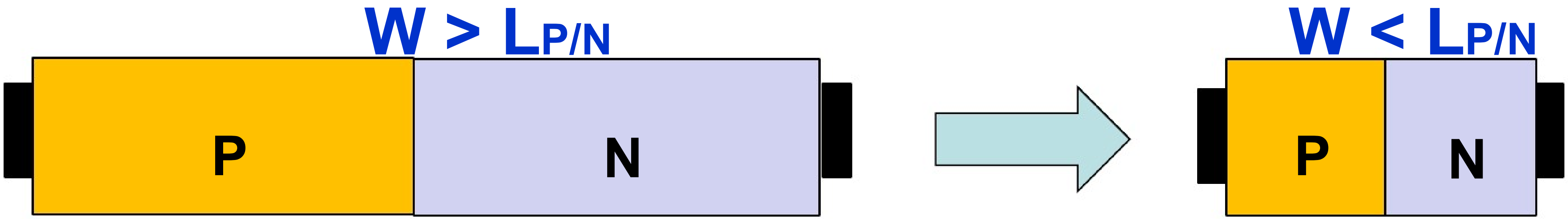
Diode – FB and RB currents

With a current limiting resistor, virtually all the voltage drops across the resistor.

Without a current limiting resistor, the bands flatten out, enormous currents flow and the diode is destroyed.



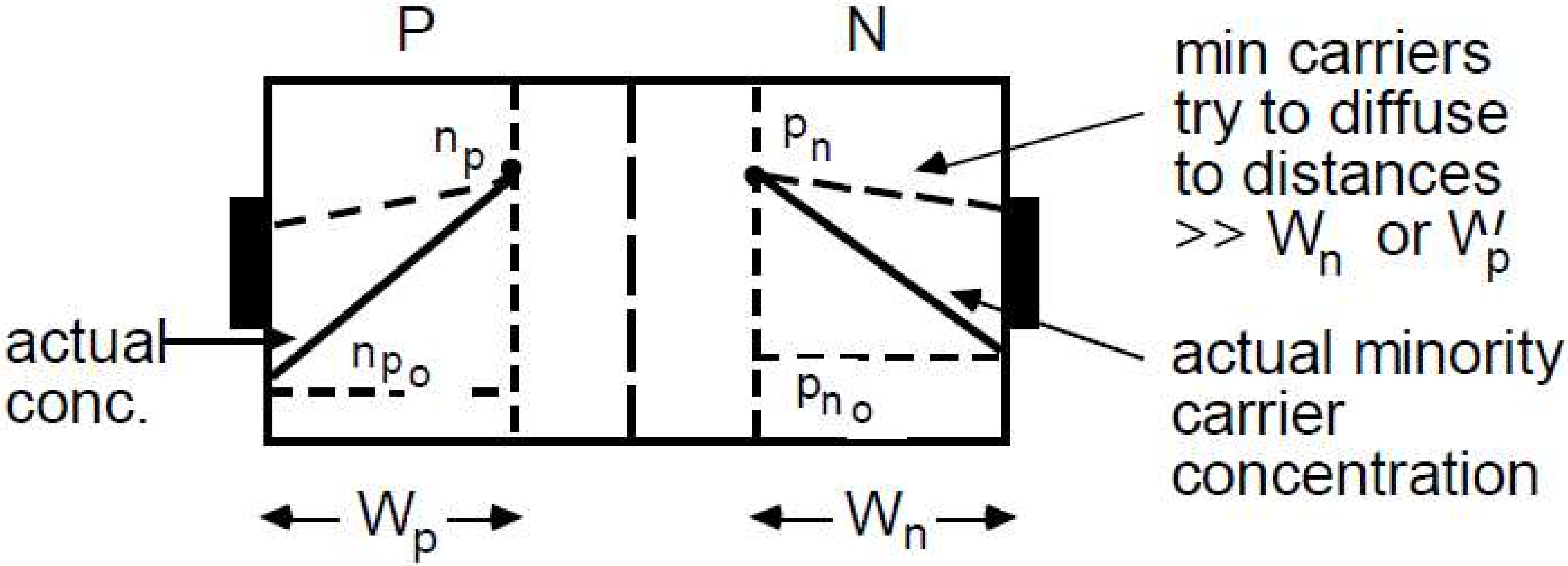
Short Base Diode



A second limiting case of considerable practical importance occurs when the length of the "neutral" regions are << L_p and/or L_n, the minority carrier diffusion lengths. A particularly important example is the base region of the bipolar transistor. The surface recombination velocity at the Ohmic contacts is ≈ ∞ because of the large number of traps available there.

Hence any minority carrier which reaches an Ohmic contact instantly recombines and the minority carrier density goes to p_n or n_p, the thermal equilibrium values of n_i^2/N_D or n_i^2/N_A.

Short Base Diode Carrier Distributions



Hence, the minority carrier boundary conditions imposed by the Ohmic contacts are:

$$p_n(x = W_n) = p_{no}$$

$$n_p(x = -W_p) = n_{po}$$

Short Base Diode Current Voltage Characteristic

If we assume **negligible recombination** occurs between the depletion region and Ohmic contacts, but entirely at W_n and $-W_p$, then the minority carrier currents (due to diffusion) are constant and are given by

$$J_P = qD_P \frac{dp_n}{dx} = qD_P \frac{\Delta p_n}{W_n}$$

$$= qD_P \frac{p_{no}}{W_n}\left(e^{qV_A/kT} - 1\right)$$

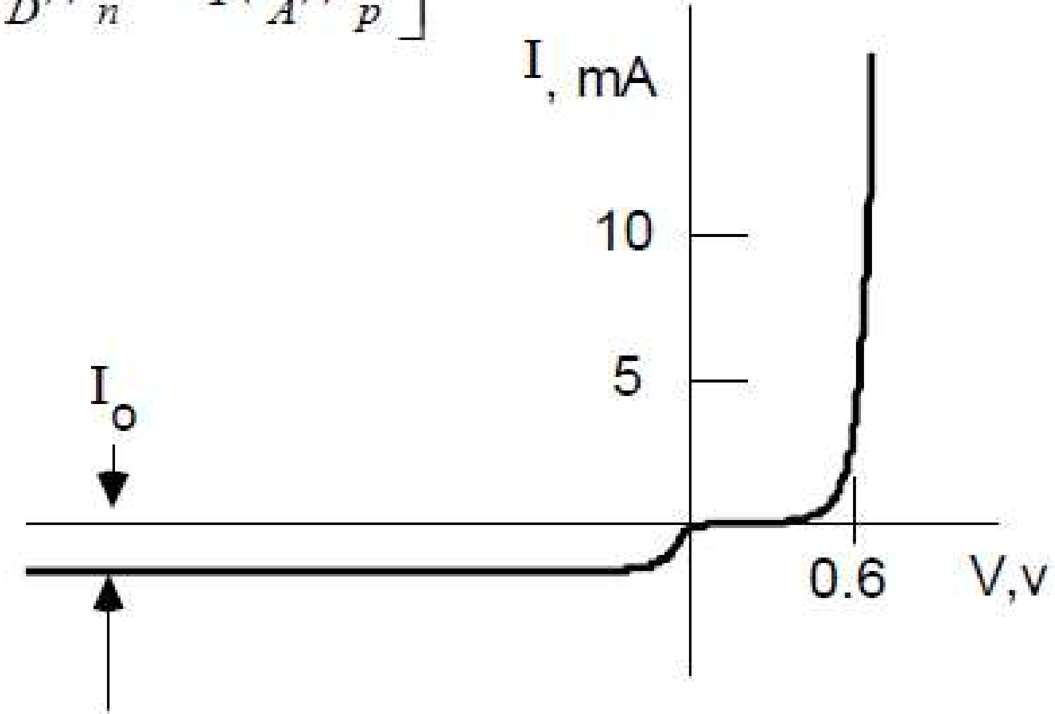
$$= qD_P \frac{n_i^2}{N_D W_n}\left(e^{qV_A/kT} - 1\right) \qquad (32)$$

$$J_N = qD_N \frac{n_i^2}{N_A W_p}\left(e^{qV_A/kT} - 1\right) \qquad (33)$$

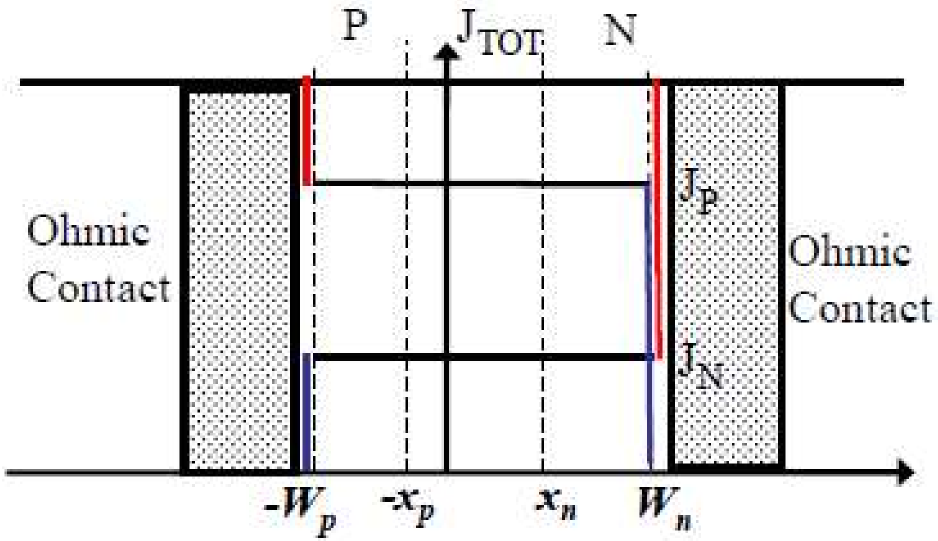
Short Base Diode Current Voltage Characteristic

No majority carrier recombination currents flow because there is no recombination. The total current is again given by the sum of the two minority diffusion currents,

$$I = qAn_i^2\left[\frac{D_P}{N_D W_n} + \frac{D_N}{N_A W_p}\right]\left(e^{qV_A/kT} - 1\right) \tag{34}$$

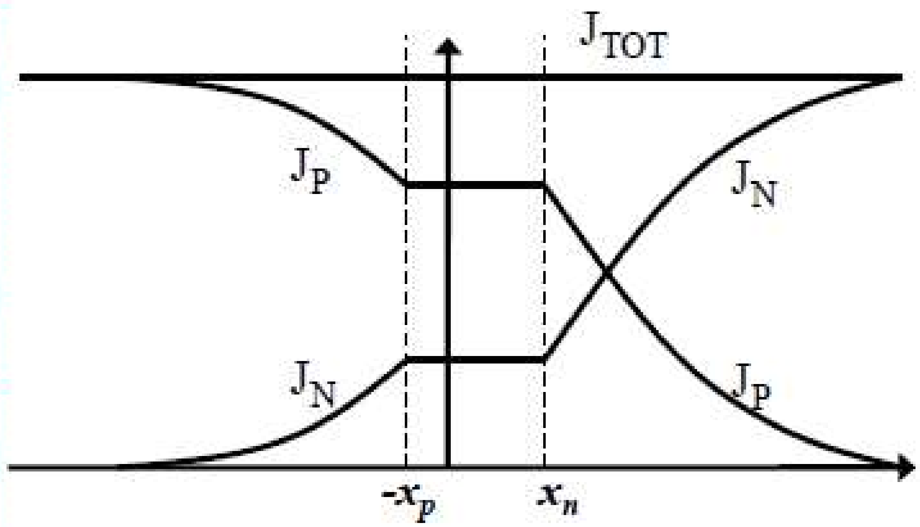


Comparison of Short vs Long Base Diode



$$I = qAn_i^2\left[\frac{D_P}{N_D W_n} + \frac{D_N}{N_A W_p}\right]\left(e^{qV_A/kT} - 1\right)$$

No recombination in quasi-neutral region (J_N & J_P constant)



$$I = qAn_i^2\left(\frac{D_P}{N_D L_P} + \frac{D_N}{N_A L_N}\right)\left(e^{qV_A/kT} - 1\right)$$

Recombination in quasi-neutral region to reach p_{n0} and n_{p0}

Only difference is diode length rather than diffusion length. The diodes are indistinguishable and both can be represented by the expression,

$$I = I_o\left(e^{qV_A/kT} - 1\right)$$

Comparison of Short and Long Base Diode

Comparing the long and short base diodes, equations (30) and (34) are identical except for the characteristic length appropriate to the geometry ($L_{P,N}$ or $W_{p,n}$). Actual diodes may represent intermediate cases. In either case, it is the diffusion and recombination of minority carriers on the lightly doped side of the junction which largely determines the current.

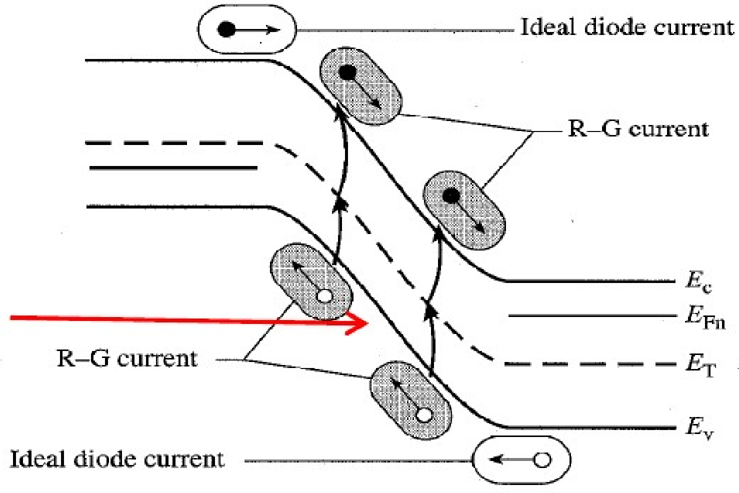
Non-ideal PN diode I-V characteristics

Space Charge Recombination Currents

We have assumed to this point that all the minority carriers make it across the depletion region. This is often not true in practice, as there will be some generation and recombination through trapping centers. Under reverse bias, these centers act as generation sites, increasing I_o over the ideal value calculated earlier and increasing with bias voltage as the depletion region gets wider. Under forward bias, traps act as recombination sites, increasing the forward current, particularly at low forward bias, but the number of traps is fixed, thus at higher biases, they are saturated and the current becomes more “ideal”.

Generation and Recombination in the Depletion Region

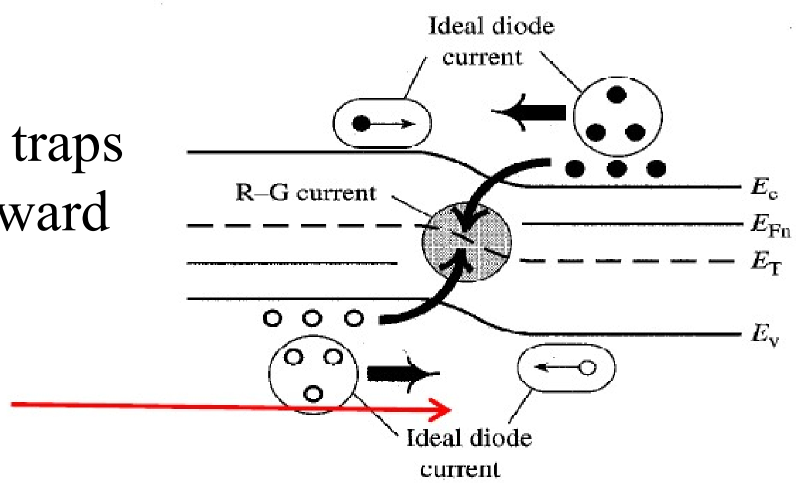
Number of active traps increases with reverse bias



(a) Reverse bias

Photo detector operates in reverse bias: generation current

Number of active traps saturates with forward bias



(b) Forward bias

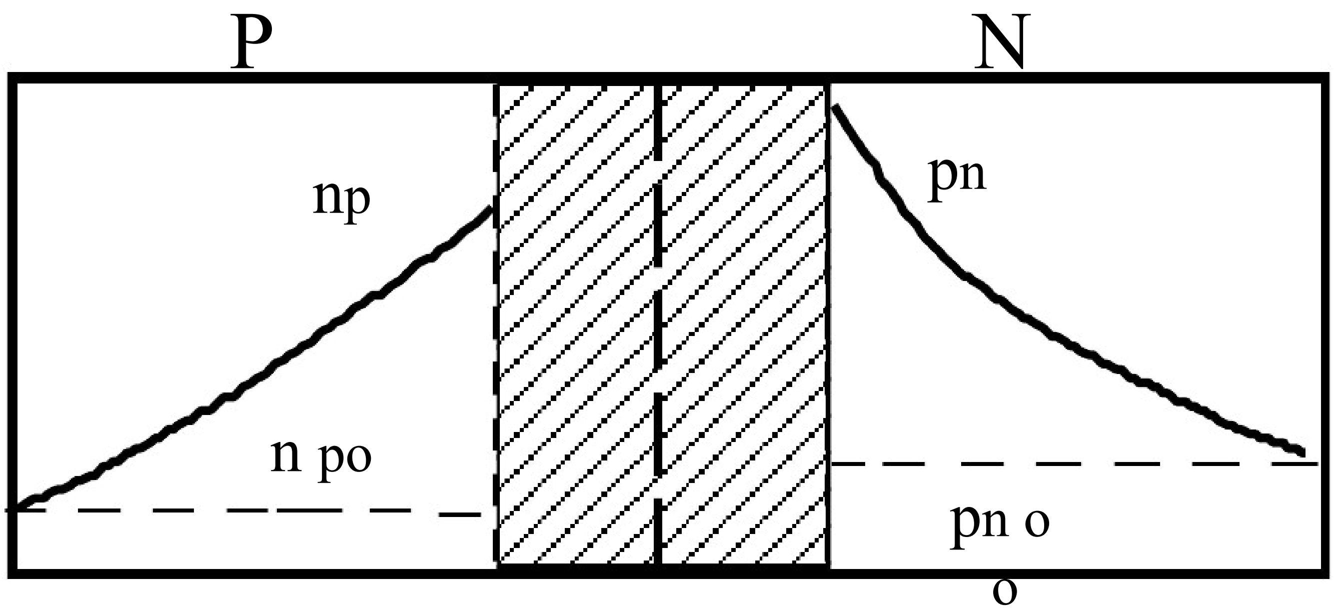
Generation and Recombination act as **added parallel paths** to the ideal diode current

Space Charge Recombination and Generation Currents

The total diode current is given by:

$$I = I_{DIFF} + I_{R\text{-}G}$$

Where I_{DIFF} is the current given by the ideal diode equations (30) or (34) and $I_{R\text{-}G}$ is the generation or recombination current.



Space Charge Recombination Current

Eq. from previous lecture gives us the SRH recombination expression:

$$R = \frac{C_n C_p N_t (np - n_i^2)}{C_n(n + n') + C_p(p + p')}$$

- Within the depletion region, n=p=0

$$R = \frac{-C_n C_p N_t n_i^2}{C_n n' + C_p p'}$$

- If trap locate at the intrinsic Fermi level, then n'=p'=ni

$$R = \frac{-n_i}{\frac{1}{N_t C_p} + \frac{1}{N_t C_n}} = \frac{-n_i}{\tau_{p0} + \tau_{n0}} = \frac{-n_i}{2\tau_0} \equiv -G$$

Space Charge Recombination Current: reverse bias

The generation current density may be determined from

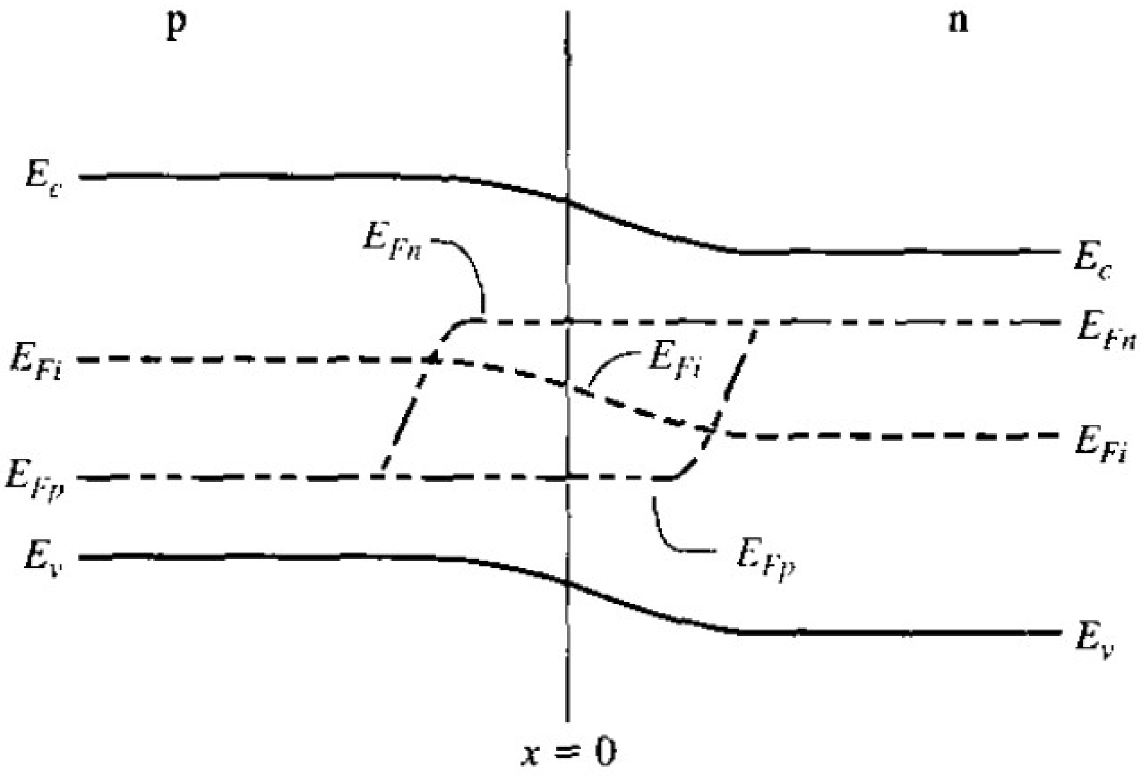
$$J_{gen} = \int_0^W eG\,dx$$

$$\boxed{J_{gen} = \frac{en_i W}{2\tau_0}}$$

Which is NOT constant because $W \propto V^{1/2}$

$$J_R = J_s + J_{gen}$$

Space Charge Recombination Current: forward bias



$$n = n_i \exp\left[\frac{E_{Fn} - E_{Fi}}{kT}\right]$$

$$p = n_i \exp\left[\frac{E_{Fi} - E_{Fp}}{kT}\right]$$

$$E_{Fn} - E_{Fi} = E_{Fi} - E_{Fp} = \frac{eV_a}{2}$$

$$n = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

$$p = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

Space Charge Recombination Current: forward bias

$$R = \frac{np - n_i^2}{\tau_{p0}(n + n') + \tau_{n0}(p + p')}$$

$$n = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

$$p = n_i \exp\left(\frac{eV_a}{2kT}\right)$$

assume that $n' = p' = n_i$ and that $\tau_{n0} = \tau_{p0} = \tau_0$

$$R_{\max} = \frac{n_i}{2\tau_0} \frac{[\exp(eV_a/kT) - 1]}{[\exp(eV_a/2kT) + 1]} = \frac{n_i}{2\tau_0} \exp\left(\frac{eV_a}{2kT}\right)$$

Space Charge Recombination Current: forward bias

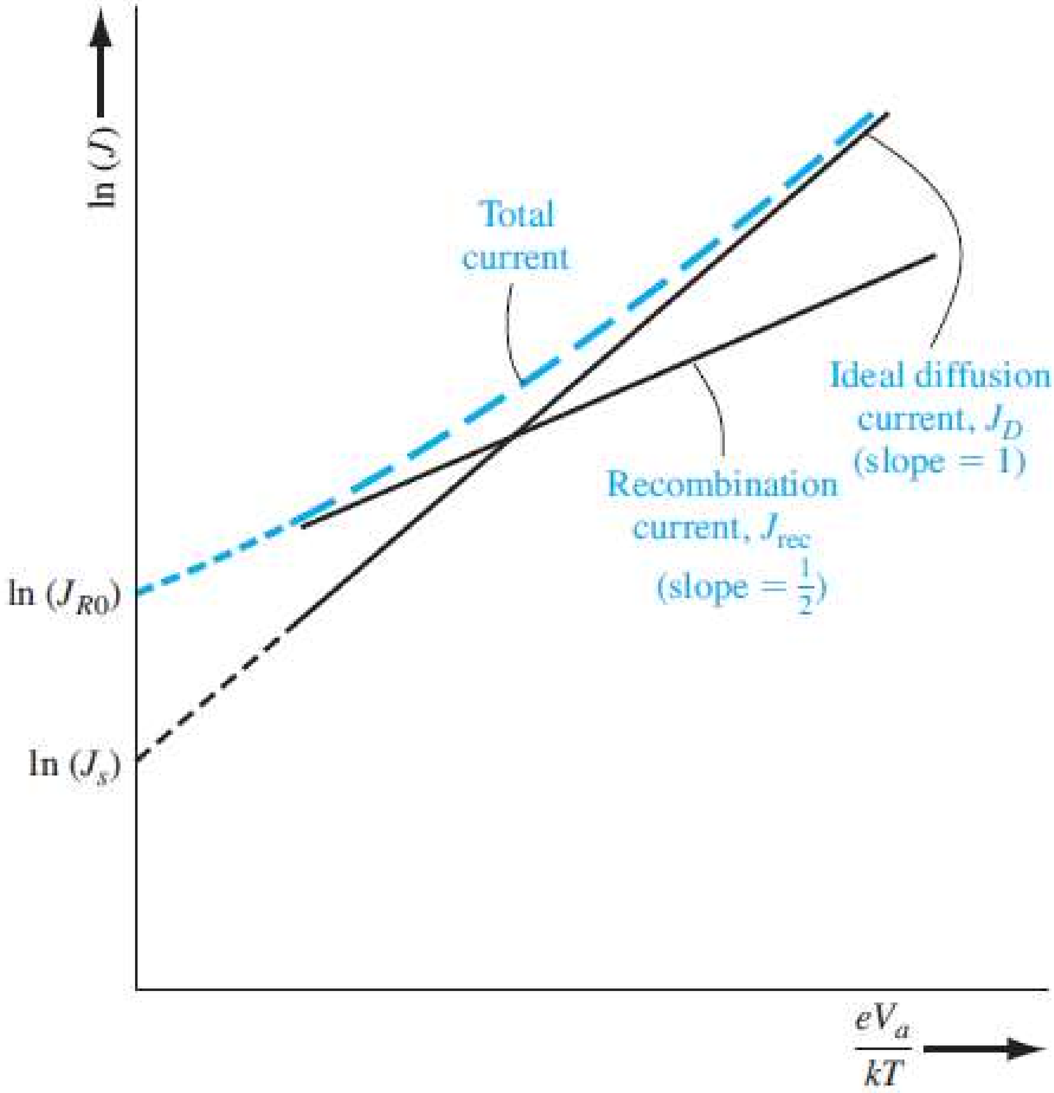
$$J_{rec} = \int_0^W eR\,dx = ex'\frac{n_i}{2\tau_0}\exp\left(\frac{eV_a}{2kT}\right)$$

- Total current is

$$J = J_{rec} + J_D = J_{r0}\exp\left(\frac{eV_a}{2kT}\right) + J_s\exp\left(\frac{eV_a}{kT}\right)$$

$$\ln J_{rec} = \ln J_{r0} + \frac{eV_a}{2kT} = \ln J_{r0} + \frac{V_a}{2V_t}$$

$$\ln J_D = \ln J_s + \frac{eV_a}{kT} = \ln J_s + \frac{V_a}{V_t}$$



The total device current now becomes

$$I = I_0\left[\exp\left(\frac{eV}{k_B T}\right) - 1\right] + I_{GR}^{o}\left[\exp\left(\frac{eV}{2k_B T}\right) - 1\right]$$

or

$$\boxed{I \cong I_S\left[\exp\left(\frac{eV}{nk_B T}\right) - 1\right] = I_S\left[\exp\left(\frac{V}{n}\frac{e}{k_B T}\right) - 1\right]}$$

where ***n*** is called the ***diode ideality factor*** or the voltage partitioning factor because the factor of 2 is a consequence of recombination occurring at the maximum recombination plane.

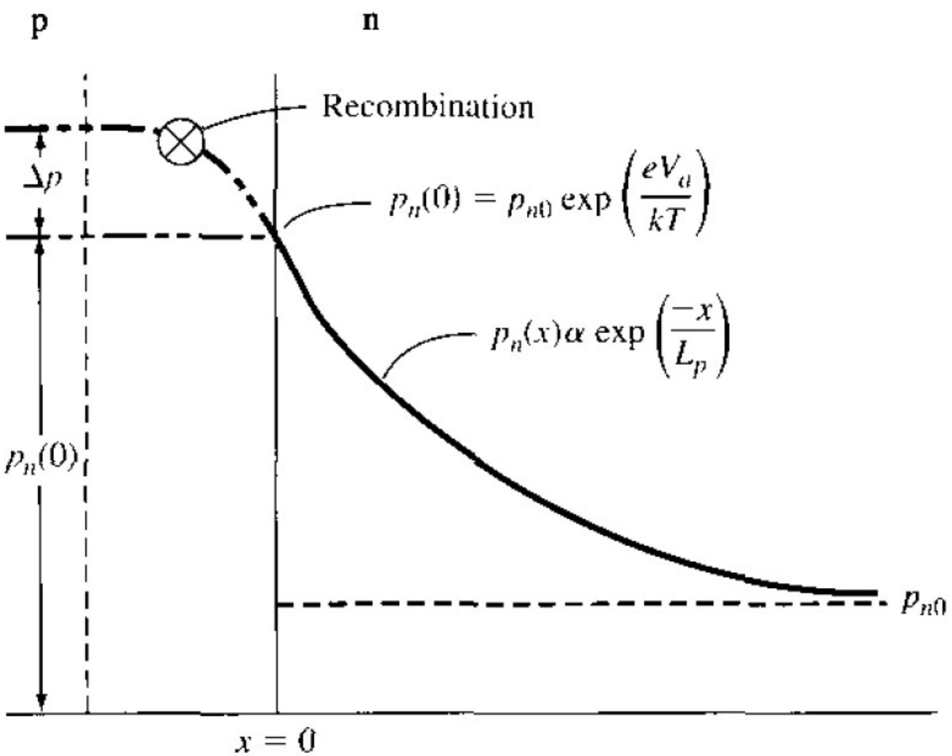
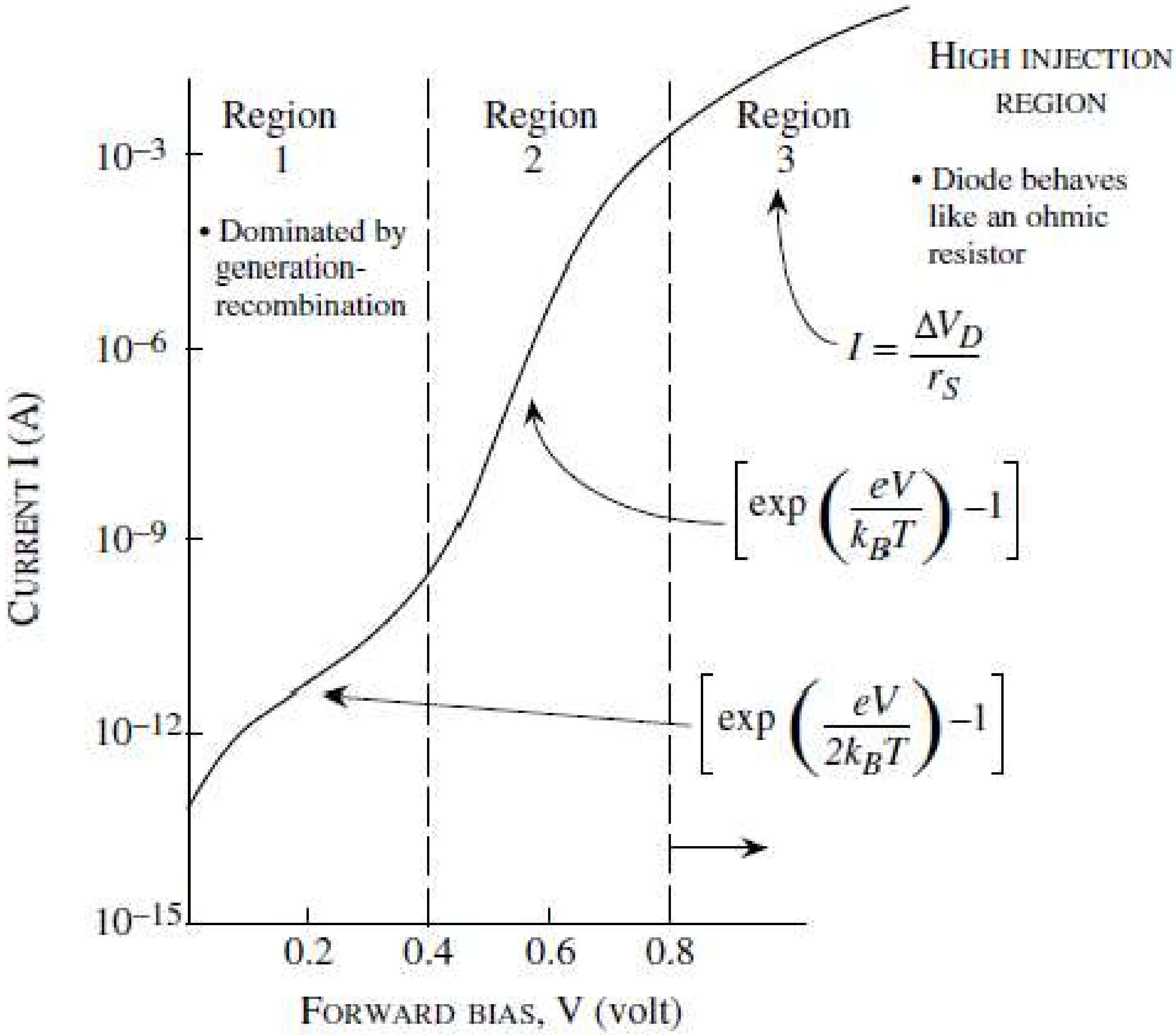


Figure 8.20 | Because of recombination, additional holes from the p region must be injected into the space charge region to establish the minority carrier hole concentration in the n region.

A log plot of the diode current in forward bias. At low biases, the recombination effects are quite pronounced, while at higher biases the slope becomes closer to unity. At still higher biases the behavior becomes more ohmic.



1. Under forward bias, I_{rec} will dominate at small V_A but will be swamped out by the ideal diode current at high *I.*

2. Under reverse bias, I_{gen} typically dominates I_o, which is a constant for the ideal diode. I_{gen} increases as $V^{1/2}$ with reverse bias, the same dependence as W.

High-Voltage Effects in Diodes

In deriving the current-voltage relation we have made two important assumptions:

i) the excess carrier density injected across the depletion region is small compared to the majority charge density;

ii) the reverse current saturates since it is due to the carriers drifting across the depletion region and is limited by the diffusive flux of minority carriers to the junction

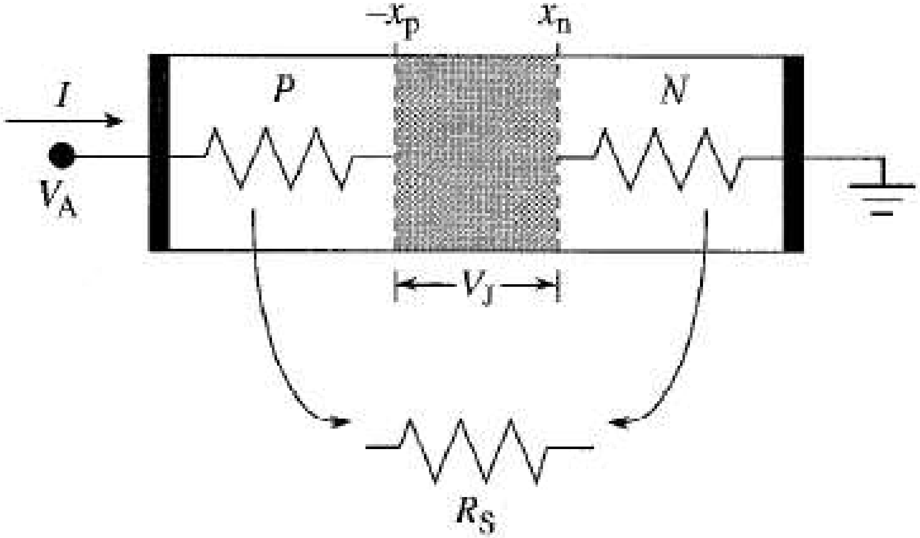
Forward Bias: High Injection Region

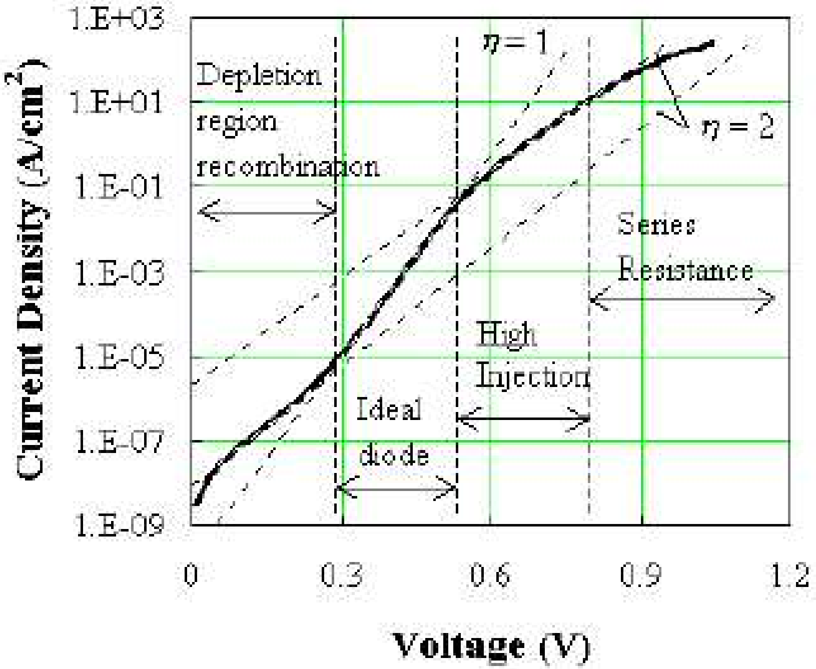
We have so far assumed that the injection density of minority carriers was low so that the voltage all dropped across the depletion region. However, as the **forward bias is increased**, the injection level increases and eventually the **injected minority carrier density becomes comparable to the majority carrier density**.

When this happens, an increasingly larger fraction of the external bias drops across the undepleted region. The diode current will then saturate. The minority carriers transport is not only due to diffusion, but also due to the electric field that is now present in the **undepleted region**.

As the forward bias increases, the devices start to **behave like a resistor**, where the current-voltage relation is given by a simple linear expression. The current is now controlled by the resistance of the n- and p-type regions as well as the contact resistance.

Series Resistance





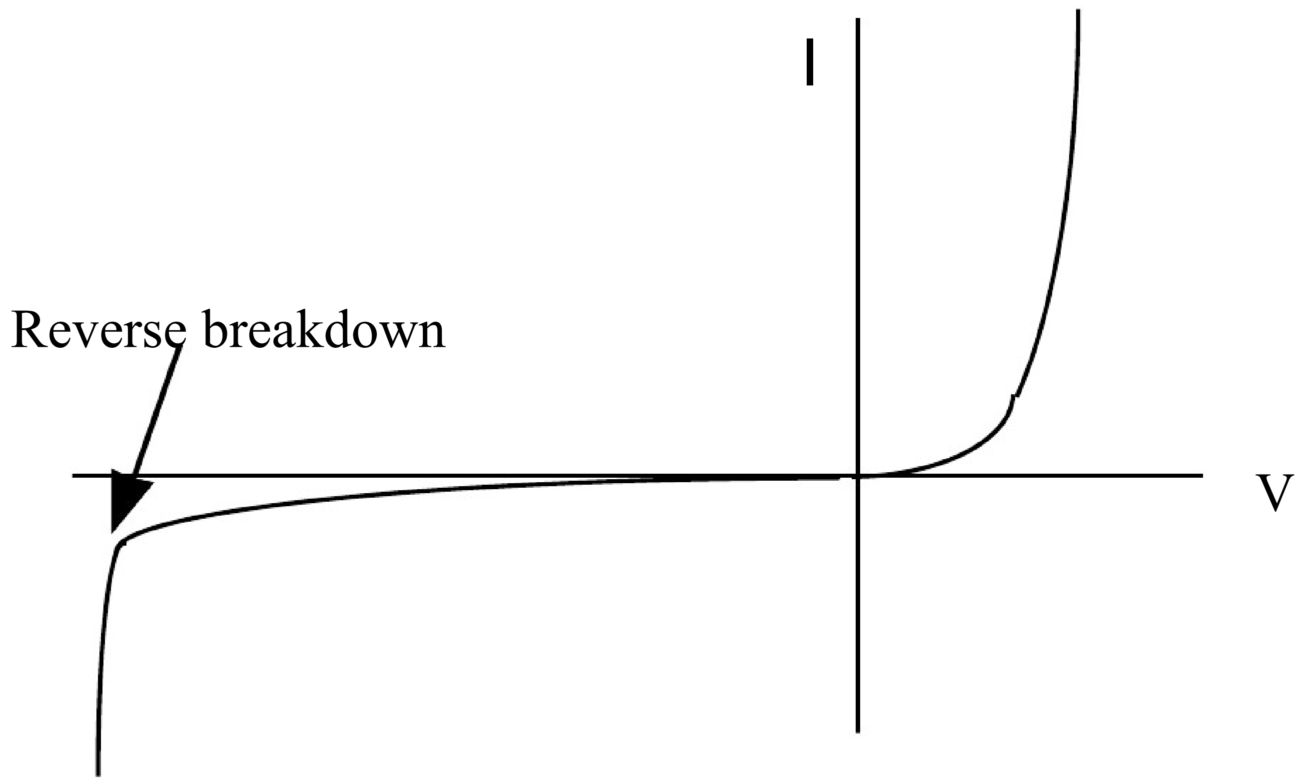
The voltage drop across the junction is reduced by the voltage drop in the neutral regions: $V_J = V_A - IR_s$

The diode current is thus modified to

$$I = I_o e^{qV_j/kt} = I_o e^{q(V_A - IR_S)/kT} \qquad (39)$$

Junction breakdown

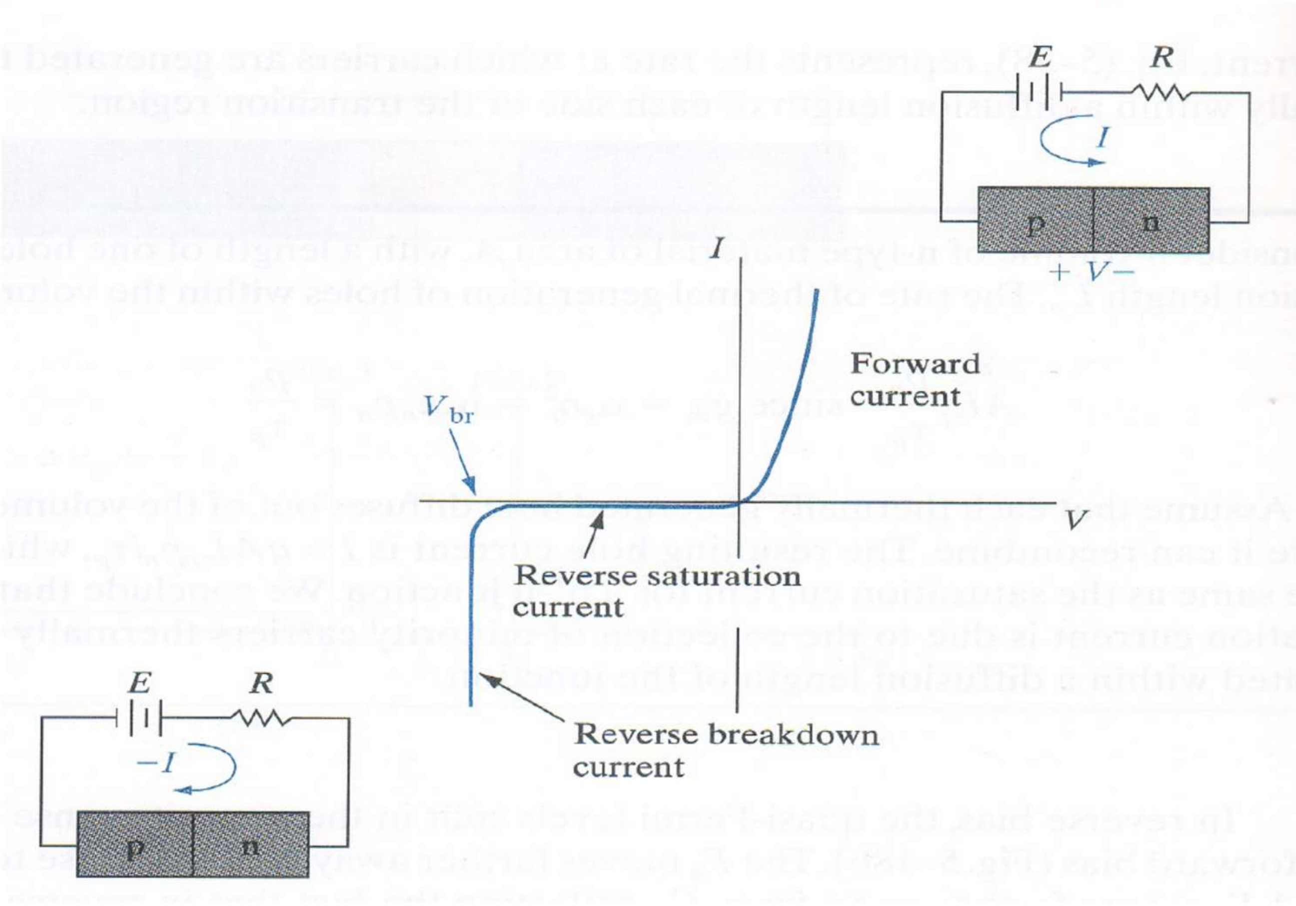
Linear Current-Voltage Characteristic



Breakdown is not explained at all by the ideal diode equations, hence represents an entirely different phenomena.

P-N junction - Reverse Bias Breakdown

In RB, the pn junction exhibits a small voltage-independent saturation current. When the RB reaches a critical voltage called the <u>breakdown voltage</u> V_{br}, the RB current increases sharply with little increase in the reverse voltage.



P-N junction - Reverse Bias Breakdown

- Reverse breakdown is not necessarily destructive if the current flow is restricted by the external circuit.
- Destruction comes only when there is an excessive reverse current flow during breakdown resulting in the overheating of the pn junction.
- The reverse current flow can be limited by incorporating an external series resistor **R**. The current will then be limited to $\mathbf{I = (E - V_{br})/R}$.

There are two different mechanisms by which reverse breakdown occurs in a pn junction, viz.,

Zener Breakdown

Avalanche Breakdown

Reverse Bias Breakdown – Zener Effect

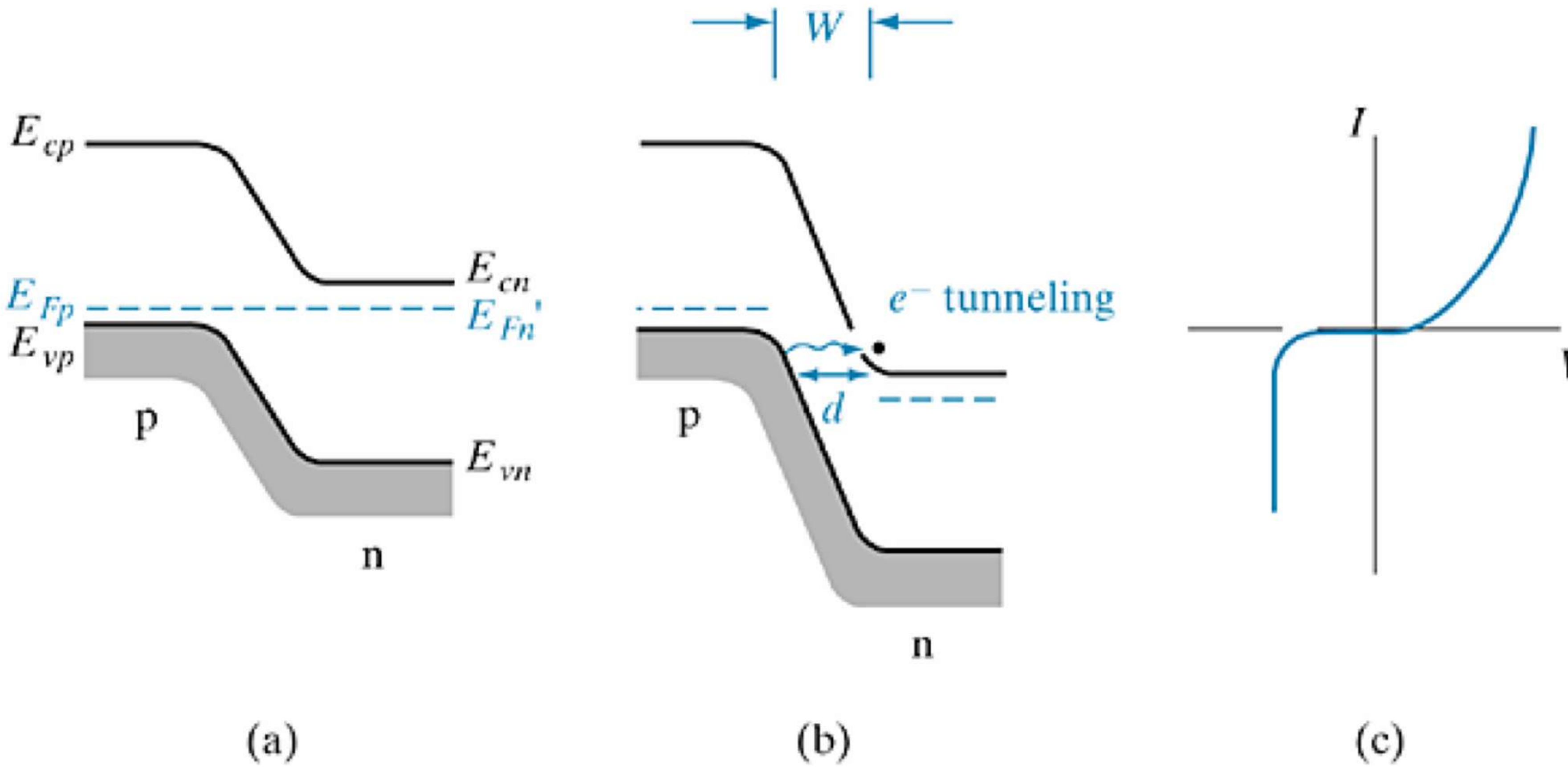
Zener Breakdown occurs in heavily doped pn junctions - more likely in abrupt junctions than diffused junctions

In heavily doped pn junctions the $\mathbf{E_{Fp}}$ is very close to $\mathbf{E_{Vp}}$ and $\mathbf{E_{Fn}}$ is very close to $\mathbf{E_{Cn}}$. Recall that at equilibrium, $\mathbf{E_{Fp}}$ and $\mathbf{E_{Fn}}$ are aligned, which implies that $\mathbf{E_{Vp}}$ will be close to $\mathbf{E_{Cn}}$.

With RB, energy barrier increases from $\mathbf{qV_o}$ to $\mathbf{q(V_o + V_r)}$, the energy band edges $\mathbf{E_{Vp}}$ and $\mathbf{E_{Cn}}$ can cross easily, viz., $\mathbf{E_{Vp} > E_{Cn}}$ (n-side CB appears opposite p-side VB).

Crossing of bands aligns large number of empty states in the n-side CB opposite filled states of p-side VB.

The electrons can tunnel across the depletion region to the empty states, the probability of which is determined by the barrier **d** separating them.

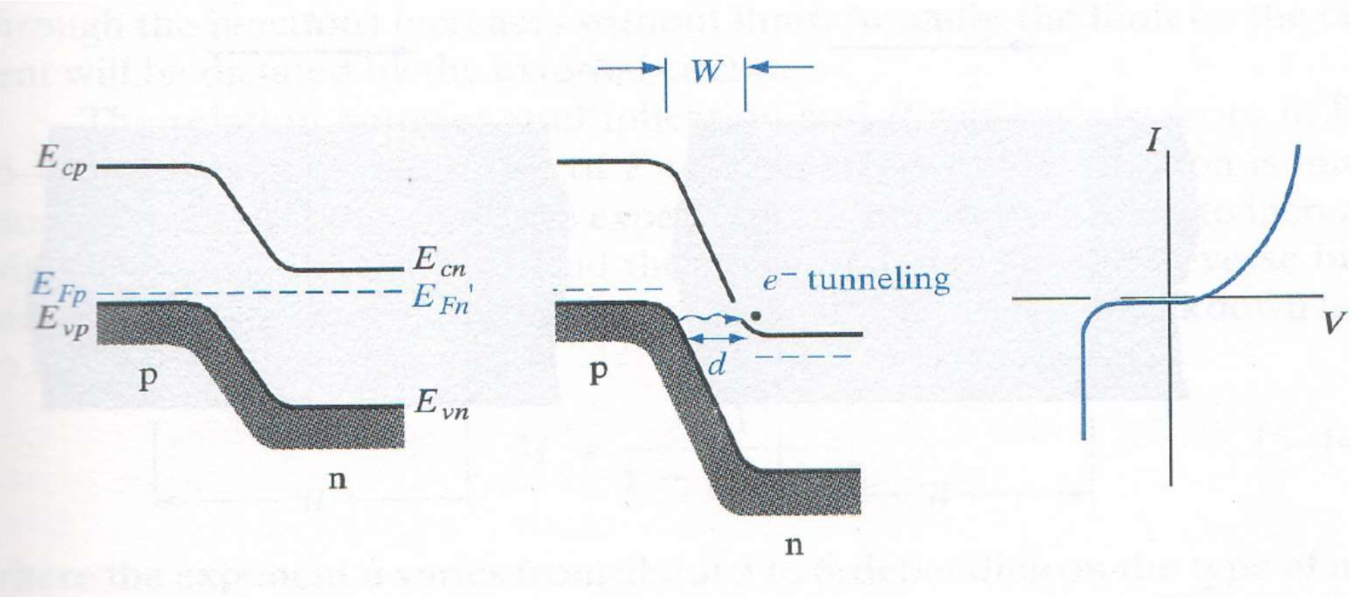


The Zener effect: (a) heavily doped junction at equilibrium; (b) reverse bias with electron tunneling from p to n; (c) *I–V* characteristic.

Reverse Bias Breakdown – Zener Effect

As V_r increases, the barrier **d** decreases, tunneling of electrons begins to occur.

Tunneling of electrons from p-side VB to n-side CB constitutes a reverse current from n to p and this phenomenon is called the Zener Effect.



(a) (b) (c)

Reverse Bias Breakdown – Zener Effect

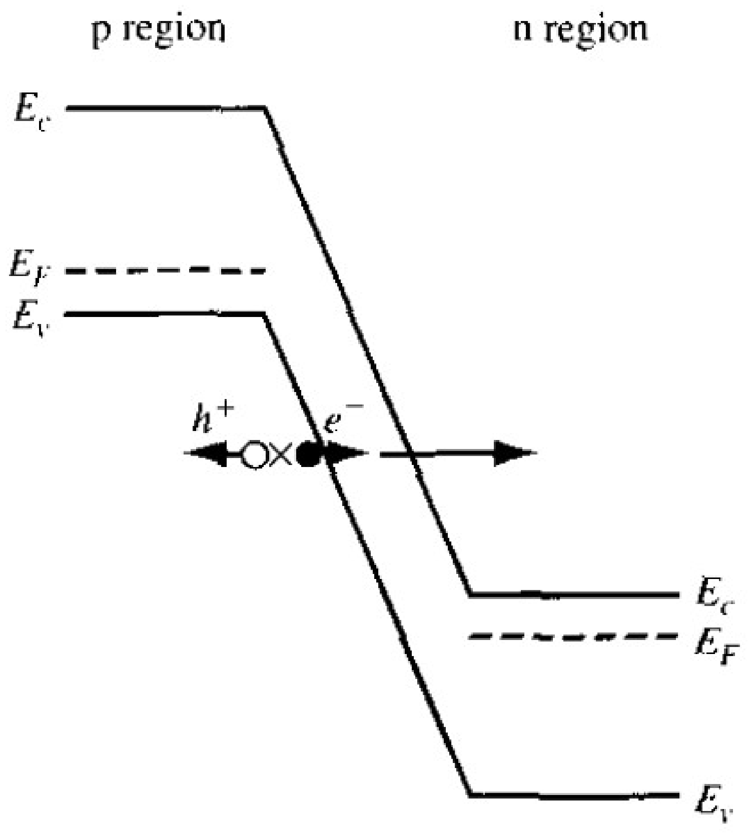
Basic requirements for tunneling current are large number of electrons separated from a large number of empty states by a narrow barrier of finite-height.

The metallurgical junction should be sharp with high doping density so that the depletion region extends only a very short distance on each side of the junction.

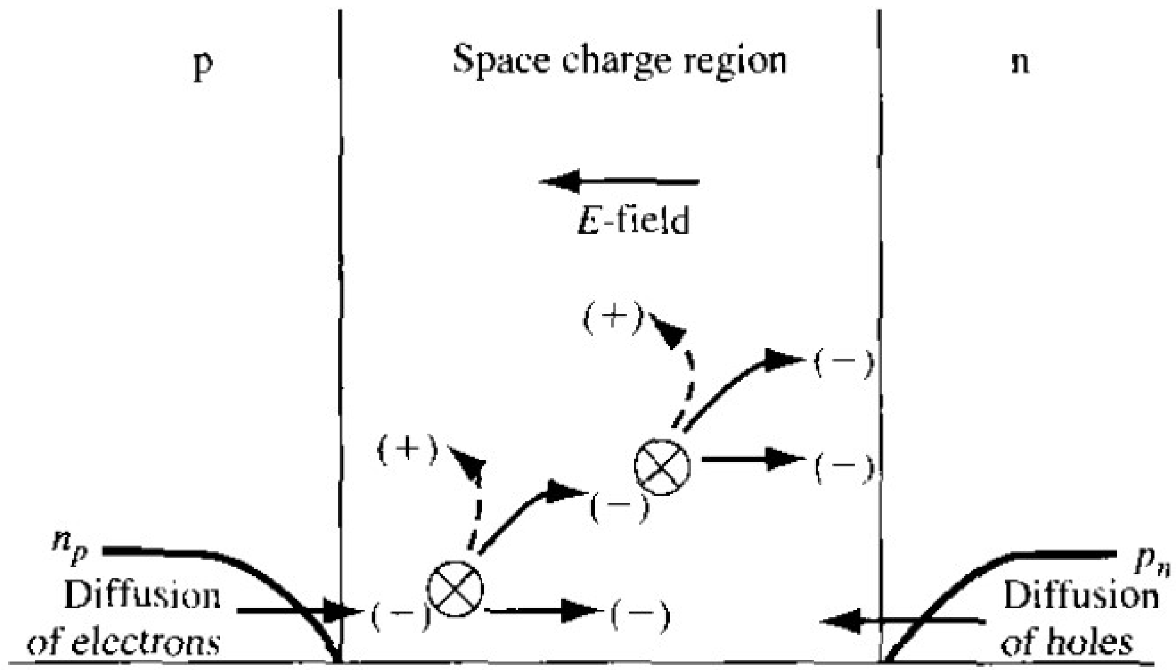
Thus, if the junction is abrupt, or if the sides of the pn junctions are heavily doped, tunneling is more likely as the transition from the p to n region in such junctions occurs over a smaller distance, implying a smaller barrier **d** for electron tunneling.

Zener breakdown usually occurs at low RB voltages, with $\mathbf{V}_{br}$ of up to several volts. For $\mathbf{V}_{br}$ exceeding this range of voltages, another breakdown mechanism, the avalanche breakdown, is responsible.

Avalanche Breakdown

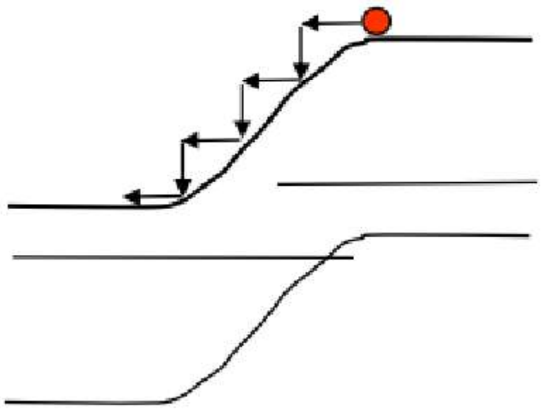


(a)

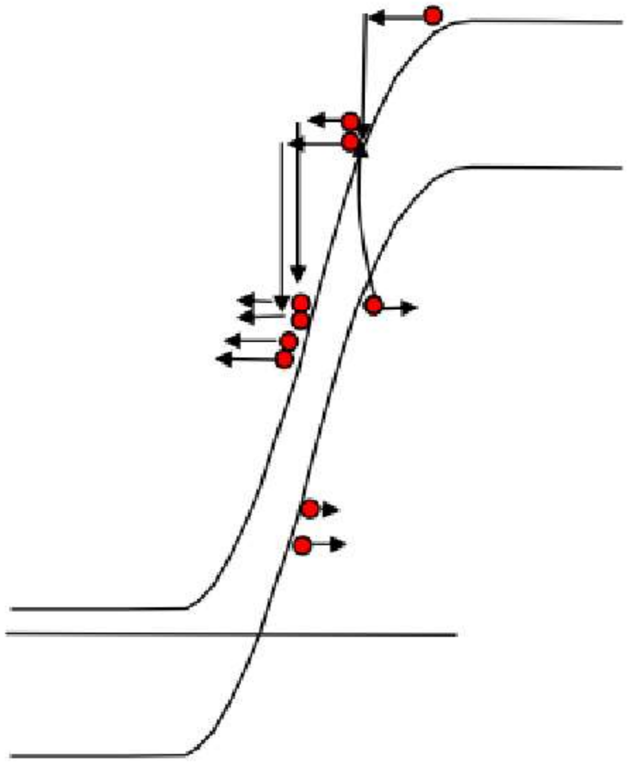


(b)

Avalanche Breakdown



(a) Small reverse bias

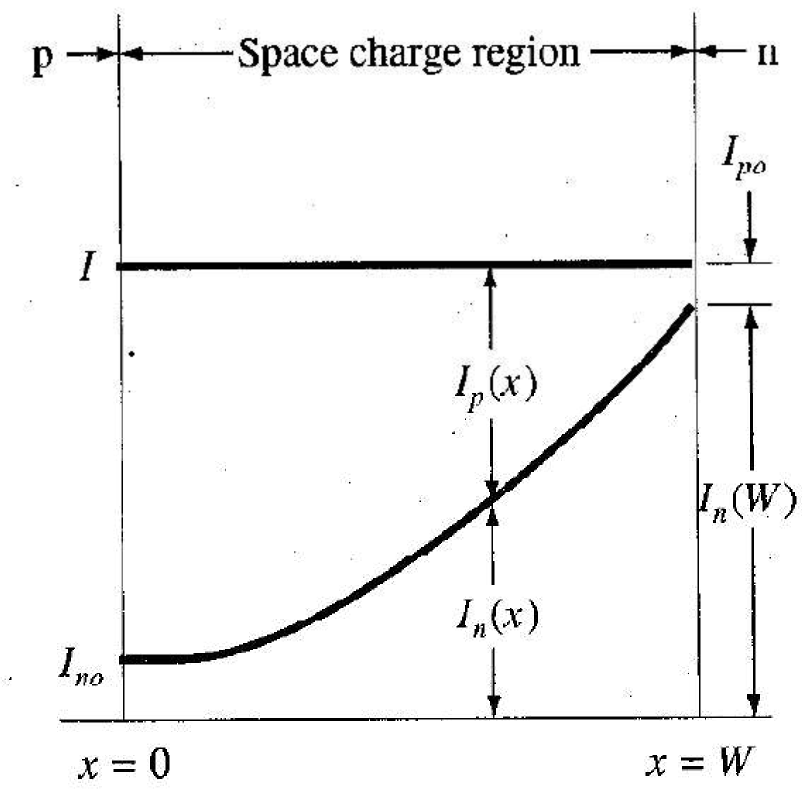


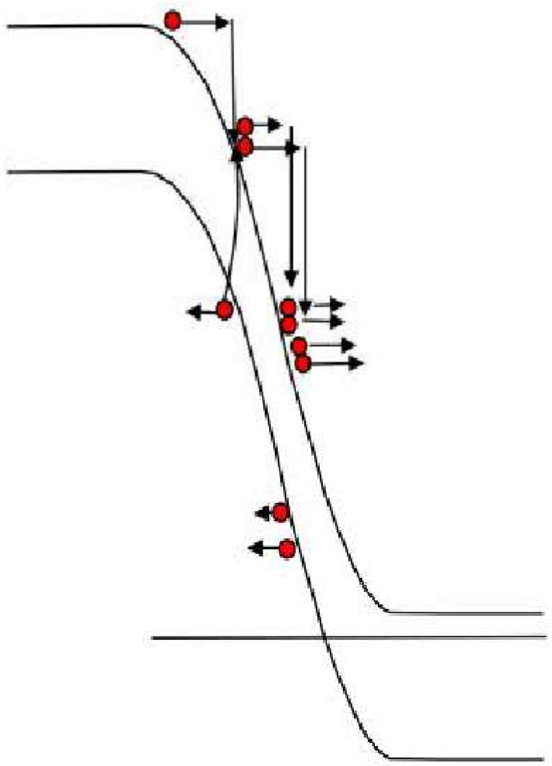
(b) Large Reverse bias -> V_{BD}

Avalanche Breakdown

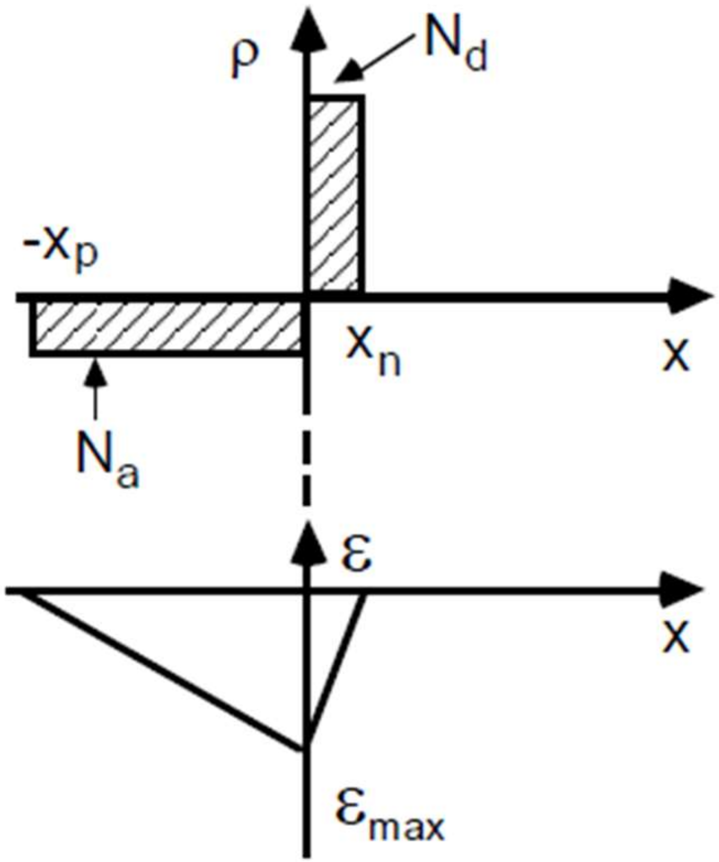
Electrons entering the depletion region (or generated there by R-G centers), gain kinetic energy under the influence of the electric field, $\mathcal{E}$ and subsequently lose it as heat through scattering. If $\mathcal{E}$ is high enough, the electron can gain sufficient energy between scattering events to break covalent bonds, freeing an electron and creating a hole. This process is repeated resulting in large reverse currents. Since the carrier must gain an energy, $E \geq E_g$, a critical field is required to initiate avalanche breakdown since at lower fields, carriers will scatter without causing impact ionization.

Electron and Hole Currents in Avalanche





Electron and hole current components through the space charge region during avalanche multiplication



Most of the depletion region occurs on the lightly doped side. From Eq. 13 and 14

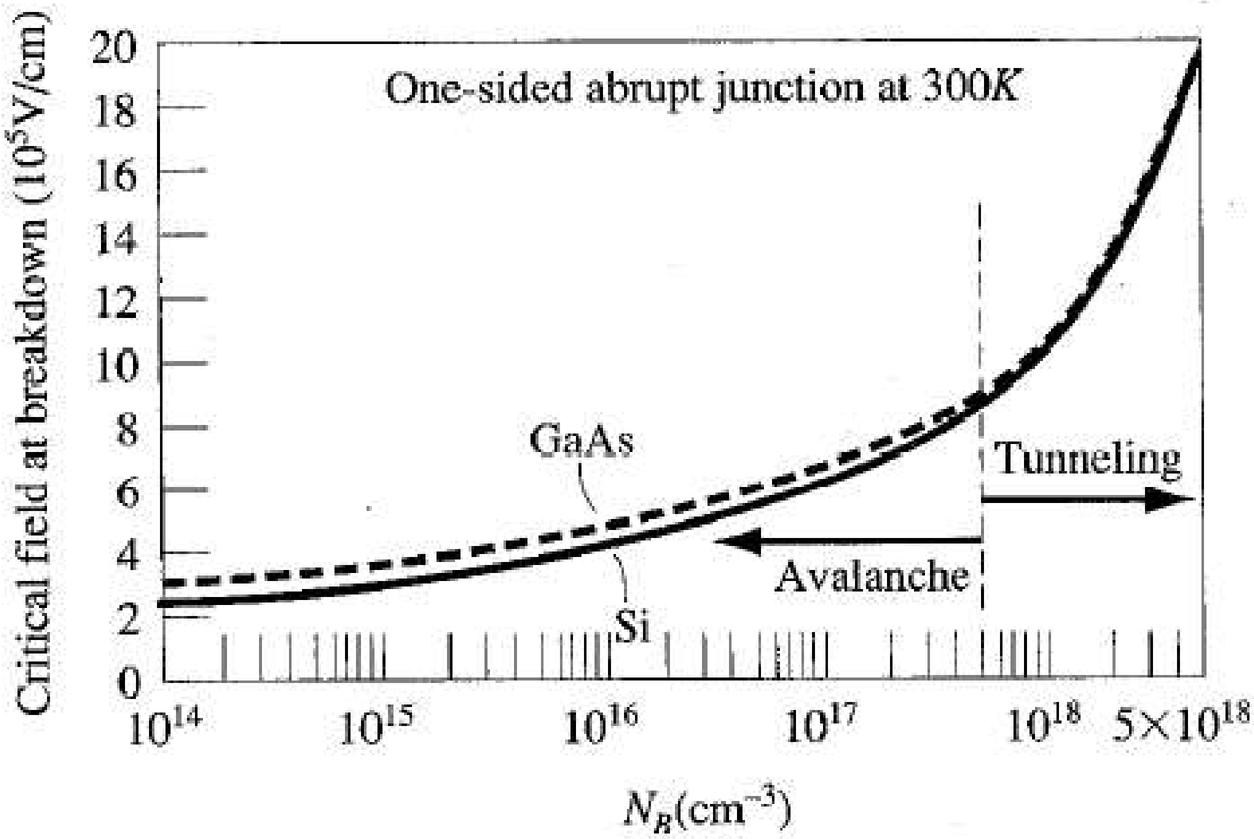
$$W \approx x_p = \left[\frac{2\varepsilon}{qN_A} \left(V_{bi} - V_A \right) \right]^{1/2}$$

$$E_{max} = \frac{eN_A x_p}{\varepsilon} = (\frac{2eN_A(V_{bi} - V_A)}{\varepsilon})^{1/2}$$

If breakdown is defined to occur when the maximum field reaches the critical field $\mathcal{E}_{CR}$, then

$$V_{BR} \approx \frac{\varepsilon \mathcal{E}_{CR}^2}{2qN_A} \qquad (52)$$

Critical Breakdown Fields in Si and GaAs



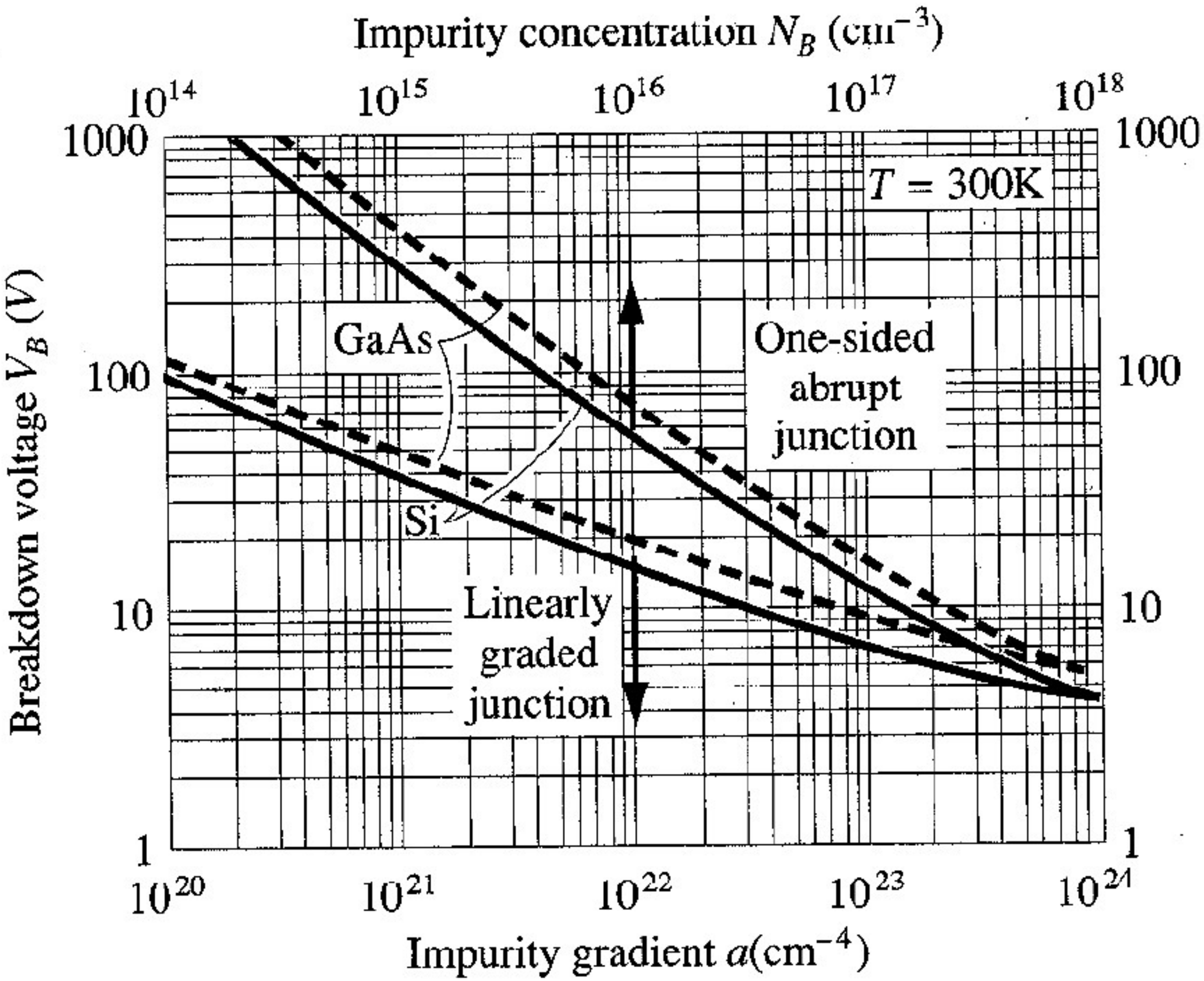
Note that the breakdown field is determined by the doping on the lightly doped side.

Note that the critical electric field, $\mathcal{E}_{CR}$, for avalanche breakdown increases moderately as N_D or N_A increases. This is because ionized impurity scattering increases and on average, slows down carriers by scattering more frequently.

Since the critical field is ≈ constant at $\mathcal{E}_{CR} \approx 3 \times 10^5$ - 8×10^5 V/cm, the breakdown voltage varies approximately inversely as the doping on the lightly doped side of the junction.

PN Junction Breakdown Voltage

Quiz: GaAs vs. Si which has higher V_{BR} ?



Note:

1. For breakdown voltage > a few volts, the inverse relationship of V_{BR} to N_A or N_D holds reasonably well.

2. At high doping levels, the V_{BR} flattens out because a different mechanism (Zener breakdown or tunneling) becomes dominant.

3. V_{BR} increases at a given N_A or N_D as the bandgap of the material increases . This makes sense because it is harder to create electron hole pairs in wider bandgap material, increasing $\mathcal{E}_{CR}$.

4.Similar calculations may be carried out for linearly graded junctions with generally similar results. Higher gradient results in lower V_{BR}.

Summary

- ☐ Ideal PN diode I-V characteristics: forward and reverse current
- ☐ Short VS long diode
- ☐ Non-ideal PN diode I-V characteristics: recombination and generation
- ☐ Junction break down mechanisms
- ☐ Mechanism for I-V curve at different voltage range